On Problems Quantum Computers Can Solve Efficiently

(Submitted at FEE2026)

29/07/2026

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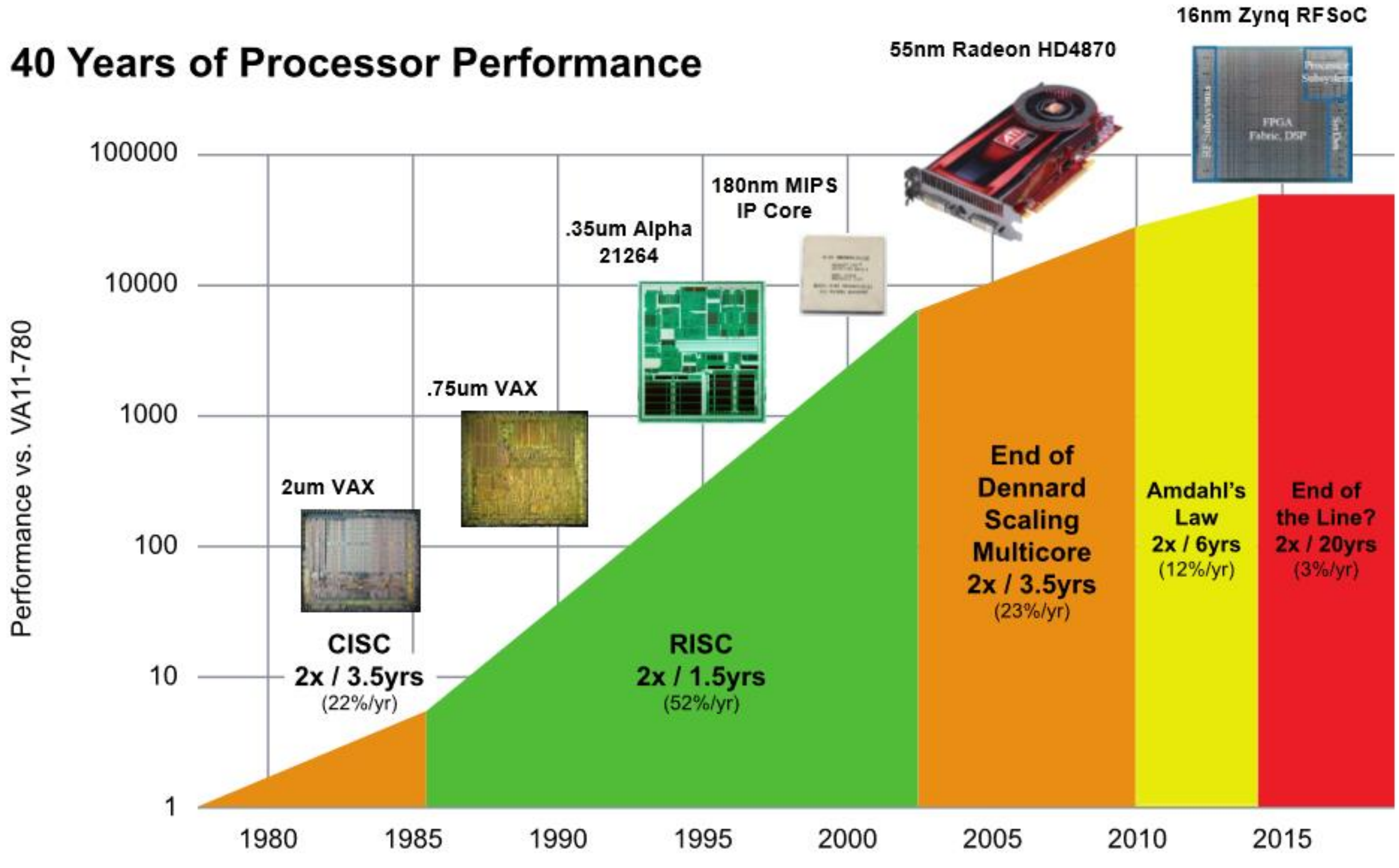
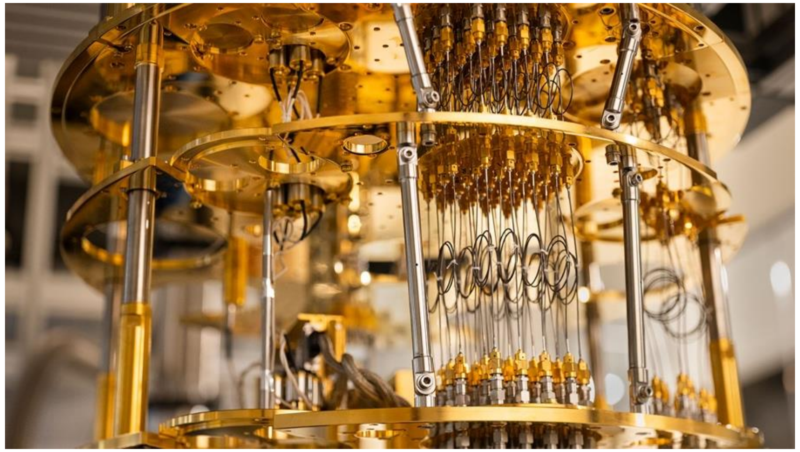


Quantum computer?



THE QUANTUM INSIDER

Quantum Computing Market Map

Non exhaustive and in no particular order. Excludes details on control systems, assembly languages, circuit design, etc.

Users
Select examples

Material Science
Merck, OTI, Airbus, BMW

Finance
Goldman Sachs, Bank of America, J.P.Morgan, Wells Fargo

Life Sciences
Odyssey Therapeutics, Boehringer Ingelheim, AstraZeneca, gsk

Other
VW, Johnson Electric, Denso

Applications
Not mapped to verticals

Software offerings
Includes control software

Not strictly categorized given diversity of operations[1]
Multiverse Computing, QC Ware, 1QBit, NQCG, Aliro Quantum, Horizon Quantum Computing, HQS Quantum Simulations, Zapata, QCI Ready-to-run Quantum Software, Entropica Labs, river lane, Strangeworks, QuantFi, Phasecraft, aq, QunaSys, Terra Quantum, ⟨b| e^{it}, ProteinQure, Keysight Technologies, Qubit pharmaceuticals, Q-CTRL, Classiq, Quantum-South, menten.AI, ParityQC

Cloud access to QPUs
aws, Google Cloud, Azure

Simulators / q-inspired / etc
Atos, Fujitsu, NVIDIA, Huawei Cloud

QPUs[2]

Superconducting
rigetti, Google AI Quantum, OQC, Quantware, IBM Q, qci, IQM, Alice&Bob, Origin Quantum, Nord Quantique, aws, seeqc, D-Wave The Quantum Computing Company, Alibaba.com

Ion Trap
Quantinuum, IonQ, Universal Quantum, AQT, oxford ionics, QUDOOR 启科量子, eleQtron

Neutral Atoms
atom computing, ColdQuanta, QuEra, Pasqal

Silicon
Silicon Quantum Computing, intel, Quantum Motion, equal1, photonic

Photonics
PsiQuantum, QC82, Xanadu, ORCA Computing, Quandela, Turing

Other
EeroQ – Electrons on helium
Quantum Brilliance – NV Diamond

Hardware / components
Select examples only – not representative of entire ecosystem

Cryogenics (includes testing)
Oxford Instruments, FormFactor, Lake Shore Cryotronics, Bluefors, Montana Instruments, ICE, Maybell

Lights and lasers
Vescent Photonics, NKT Photonics, M Squared, Coherent, Qant, Toptica Photonics, Ligentec

Other componentry (examples)
Delft Circuits, Keysight Technologies, QM Quantum Machines, Quantum Design, CryoCoax, ZNF, Qblox, QDevil, Zurich Instruments

[1] Software offerings can be further classified into SDKs, firmware / enablers, algorithms / applications, simulators etc. but many companies are offering a mixture across the stack

[2] Many QPU providers are offering full stack services (e.g. Pasqal acquired Qu&Co, Quantinuum was originally CQC prior to merger with HQS, etc.

Quantum Technology Market Map - Quantum Communication and Security

NON-EXHAUSTIVE, NO ORDER, EXCLUDES LABS

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Multiple Security Solutions

Adtran, NORMA, Chelpis, HUB, QUANTUMXCHANGE, Quintessence Labs, QCNs, orange, 中创为量子, TAQBIT LABS, QUANTUM RESISTANT CRYPTOGRAPHY, qtlabs, QSIM+, AxionTech, SANDBOXAQ, DENCRYPT, elecnor, infotecs, TKQ, Q→NU, dacoso, NXM, ISARA, PATERO, SECURE-IC, QUANTUM IMPENETRABLE, Aliro QUANTUM, ThalesAlenia Space, QuantiCor, purevpn, QINETIQ, PQSHIELD, PQSecure, SMARTS, DREAM SECURITY, Qasky, SENETAS, IDQ, CRYPTO QUANTIQUE

Post Quantum Cryptography

01 COMMUNIQUE, TAI QUANTUM, ZYY, BTQ, AMERICAN BINARY, quantropi, cyph, ABELIAN, IRONCAP, AGILEPQ, CYPHERTOP, CEW Systems Canada, QApp, QUANTUM BLOCKCHAINS, QRL, PHYSEC, NomiDio, NORMA, ARQIT, Qualcomm, QuSecure, CRYPTO EXPERTS, utimaco, XSOC CORP, cyberhive, SQE.io, WISeKey, SSH, RESQUANT, CRYPTONEXT SECURITY, ravel, HYPERSPHERE, KMS, wedge networks, CROWN STERLING, Veridify Security, QWERX, CASTLE SHIELD, QAISEC

Quantum Communication and Security Hardware

AM, Crypta Labs, CAMBRIDGE SPACE TECHNOLOGIES, S FIFTEEN INSTRUMENTS, bridgecomm, QUANTUM NUMBERS CORP, UQD, alternatio, Optalysys, QUANTUM DICE, Quantum Base, ICARUS QUANTUM, RHEA GROUP, RANDAEMON, QUANTUM CORRIDOR, ComScire, KETS PHOTONICS, EYL, Q-m6, QTEC 九州量子, SECQAI, Alea Quantum Technologies, CERNAMI, Quside, Amphenol, kpn, iQrypto, quantum emotion, RANDOM POWER, TTI, CRYPTO4A, COWEAVER

Quantum Internet

TTI, QTI, Qubit Pharmaceuticals, Quantum Bridge, LQUOM, memQ, Q LABS, Q*bird, ThalesAlenia Space, nu, Qunnect, BOHR QUANTUM

Quantum Encryption

QUANTUM INDUSTRIES, KETS, CROWN STERLING, SPEQTRAL, MATRICS2, Aegis Quantum, MagiQ TECHNOLOGIES, TOSHIBA, HEQA SECURITY, qssys, QTI, QTI, KEEQUANT, nutshell, QuBalt, VeriQloud, LUXQUANTA, Qrypt, ZTE, 问天量子, FlipsCloud, QRATE, QEYnet

*Strict distinction between these companies is challenging, so it's defined as best as possible.

Source: The Quantum Insider Intelligence Platform

Why Quantum?

Danh mục sản phẩm công nghệ chiến lược được Chính phủ ban hành 30/4/202

Nhóm 2: Nhóm các sản phẩm công nghệ tạo động lực tăng trưởng mới, công nghệ nền tảng cho tương lai, công nghệ bảo đảm tự chủ trong lĩnh vực an ninh, quốc phòng

23. Chip chuyên dụng.

24. Truyền thông lượng tử, tính toán lượng tử và cảm biến lượng tử.

25. Hệ thống khai thác, chế biến sâu và sản phẩm chế biến sâu từ khoáng sản, dầu khí và đất hiếm.

26. Hệ thống, thiết bị, dịch vụ và giải pháp công nghệ thăm dò lòng đất, biển sâu, công trình biển và năng lượng ngoài khơi.

27. Lò phản ứng hạt nhân mô-đun nhỏ (SMR).

28. Vệ tinh và chùm vệ tinh quỹ đạo thấp quan sát Trái đất.

29. Công trình xây dựng đường sắt tốc độ cao.

30. Nền tảng công nghiệp, phương tiện, thiết bị và các hệ thống tích hợp đường sắt tốc độ cao, đường sắt đô thị.

Báo Nhân Dân điện tử

Phát triển công nghệ lượng tử, đáp ứng yêu cầu tự chủ chiến lược trong giai đoạn mới

Sáng 21/5, tại Trụ sở Trung ương Đảng, Tổng Bí thư, Chủ tịch nước Tô Lâm chủ trì cuộc làm việc của Thường trực Ban Chỉ đạo Trung ương về...

2 weeks ago

Laodong.vn

Cuộc đua công nghệ lượng tử và vị trí của Việt Nam

Công nghệ lượng tử đang trở thành lĩnh vực cạnh tranh giữa các quốc gia. Việt Nam đã có những bước đi đầu tiên, nhưng cơ hội sẽ không kéo...

3 hours ago

VnExpress

'Công nghệ lượng tử là chiến lược quốc gia'

Tổng Bí thư, Chủ tịch nước Tô Lâm yêu cầu phát triển công nghệ lượng tử gắn với an ninh quốc gia, trí tuệ nhân tạo, công nghiệp bán dẫn và...

1 week ago

In Vietnam

FPT®

Quantum AI &
Cyber Security Institute

viettel
high tech

VNQUANTUM

Quapp

VNU Institute
for Quantum
Technologies

QuantumLab

Quantar Lab
Quantum algorithms and technologies for advanced research

AISeQ

i-QM
INTERDISCIPLINARY
QUANTUM MATERIALS

VNUHCM

HCMUS: Đề án mở ngành đào tạo thí điểm trình độ đại học ngành Công nghệ lượng tử (khoa Vật lý – VLKT)

FPT: chuyên ngành **Khoa học và Công nghệ Lượng tử** thuộc ngành **Công nghệ Thông tin (BIT)**

HCMUT: ngành **Kỹ thuật lượng tử**
UIT: chuyên ngành Tính toán lượng tử

Outline

1. Quantum complexity theory
2. Representative algorithms (Grover's search)
3. Sources of quantum advantage
4. Quantum machine learning bottlenecks.

1. Quantum complexity theory

Deterministic

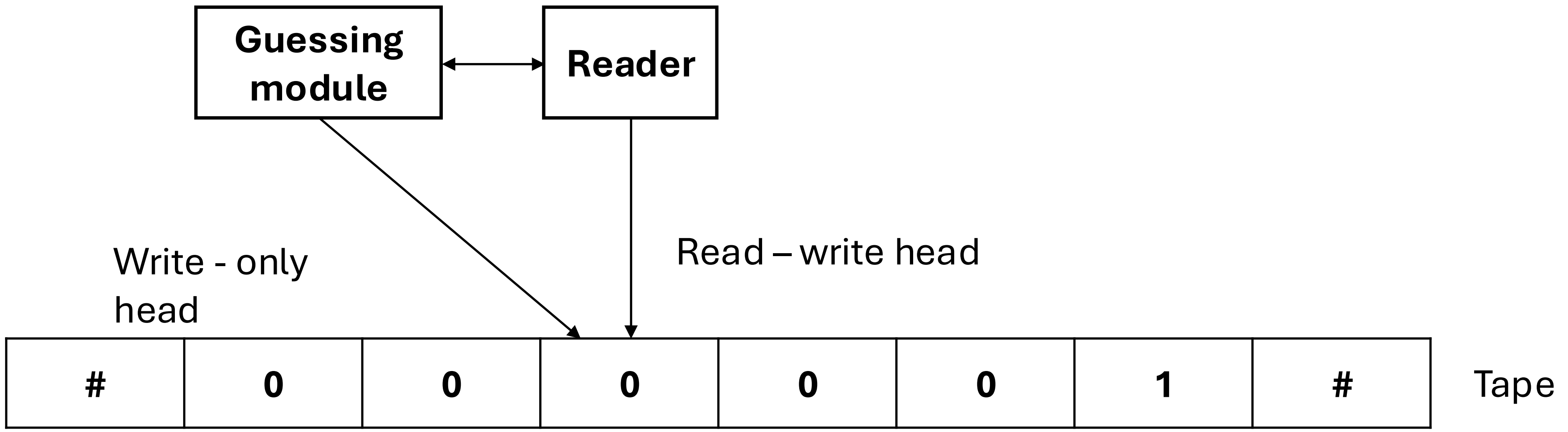
Deterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$Q = \{States\}$ $\Sigma = \{Alphabelt\} = \{0,1,\#\}$

$\delta = Q \times \Sigma \rightarrow Q \times \Sigma \times \{L, R\}$ (transition function)

δ	0	1	#
q_{start}	q_L, L	q_{accept}	q_{reject}
q_L	q_L, L	q_{accept}	q_R, R
q_R	q_R, R	q_{accept}	q_{reject}

Nondeterministic

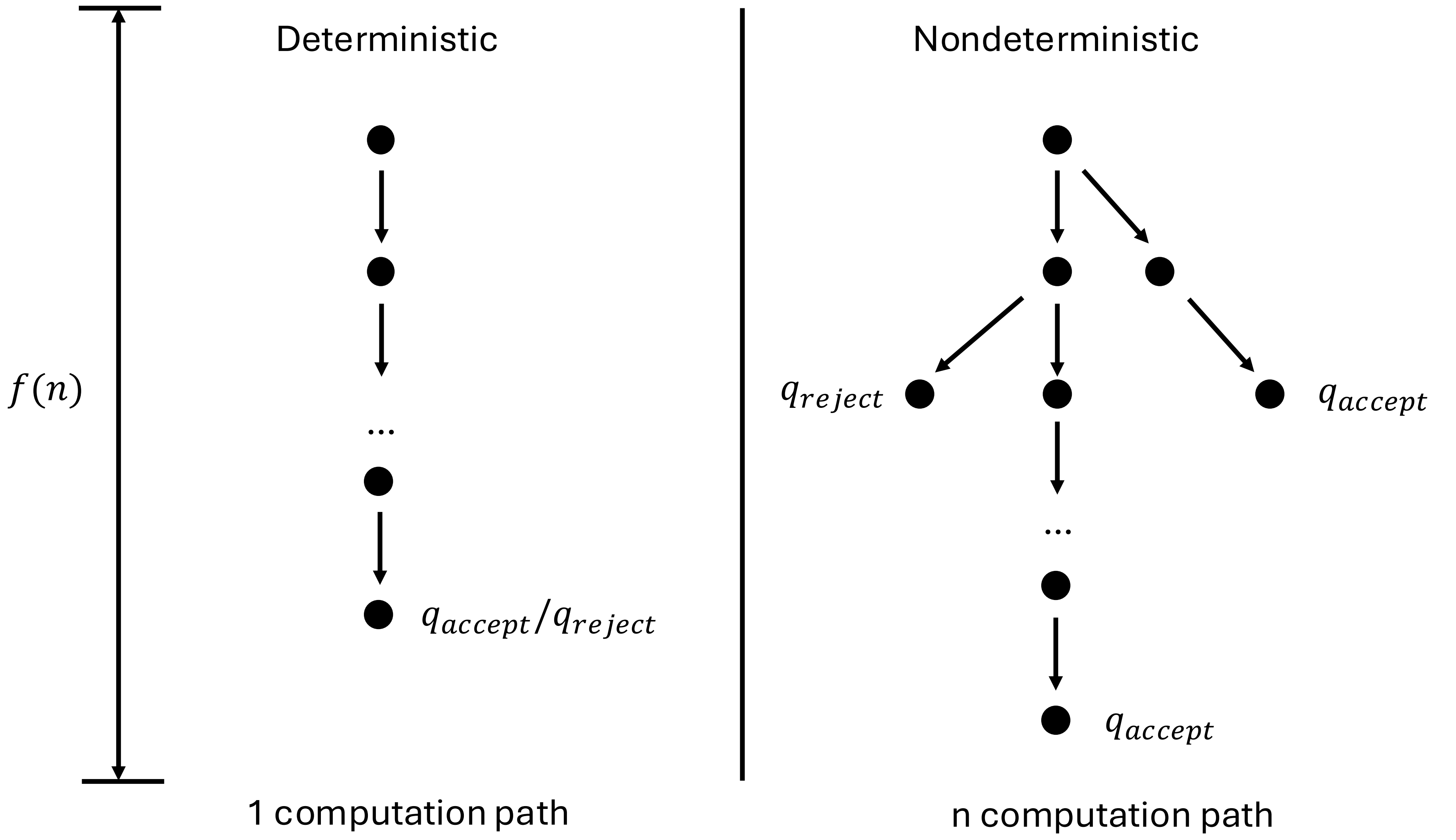


Nondeterministic

Nondeterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$\delta = Q \times \Sigma \rightarrow \mathcal{P}(Q \times \Sigma \times \{L, R\})$, $\mathcal{P}(S)$ denotes for power set of S

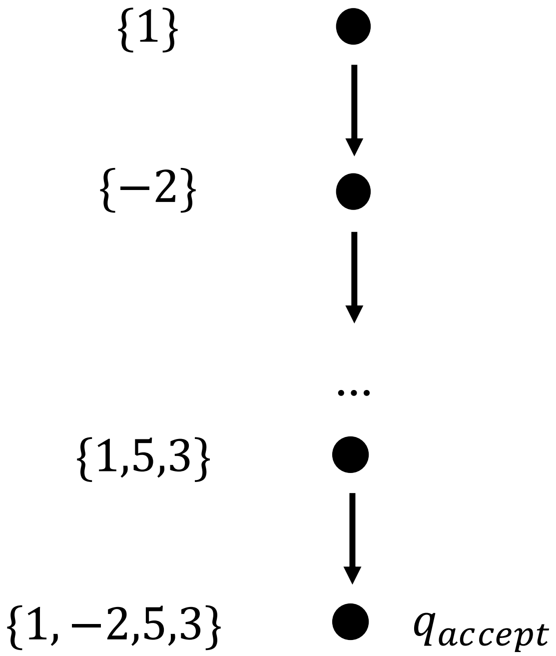
If the problem is solvable in polynomial time by a non-deterministic Turing machine, the problem belongs to NP class


$f(n)$
Deterministic
Nondeterministic
q_{accept}/q_{reject}
q_{reject}
q_{accept}
q_{accept}
1 computation path
n computation path

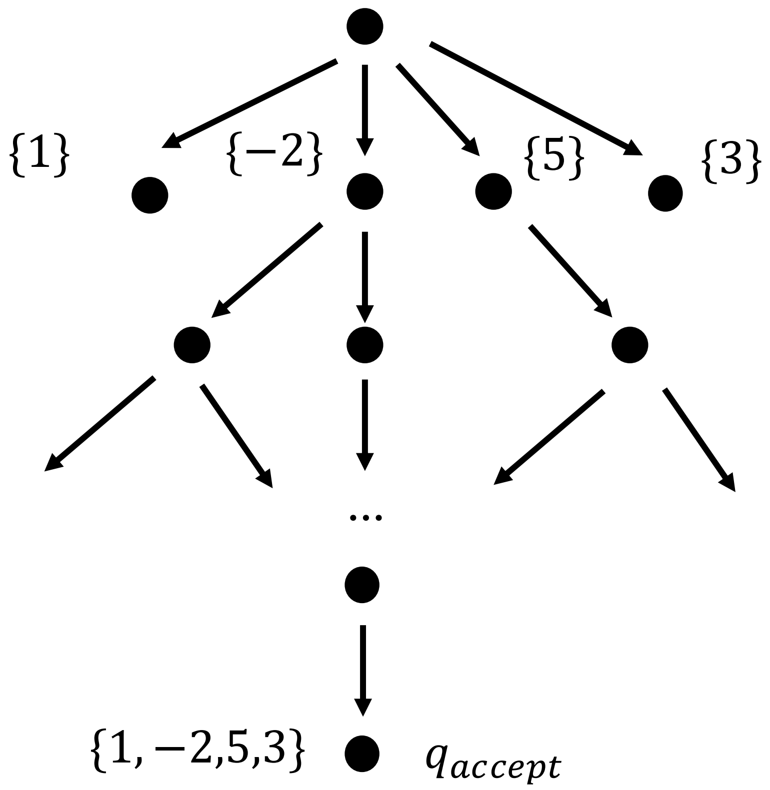
Example: Sub-set sum probel

Given a set S = $\{1, -2, 5, 3\}$. What sub-set of S has sum equal 7?

Deterministic

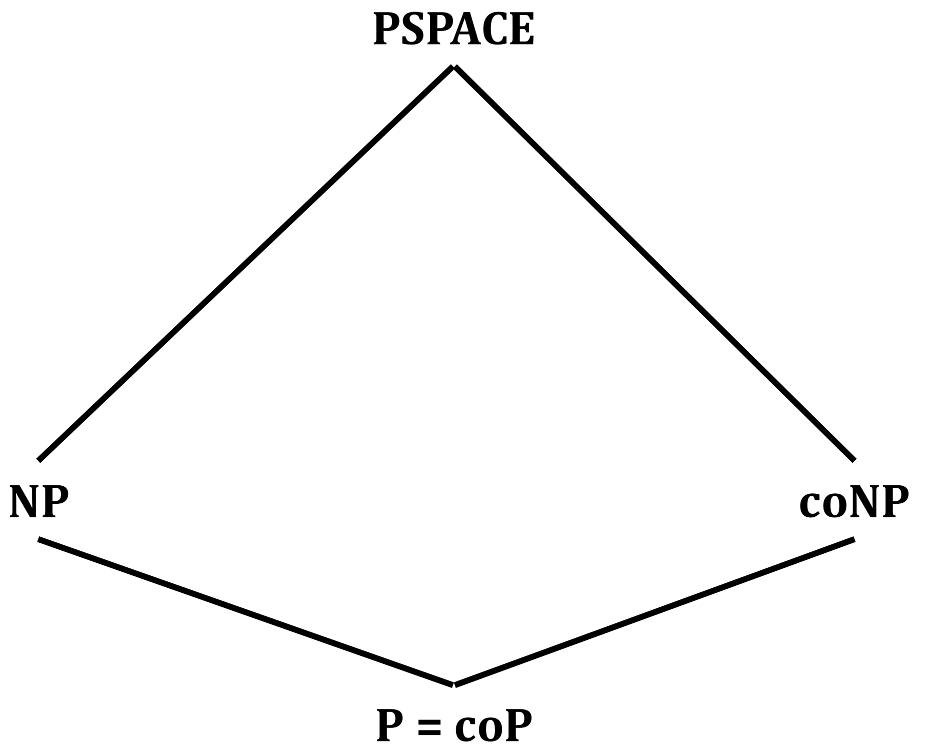


Nondeterministic



S = $\{1, -2, 5{,}3\}$.

Polynomial number of **SPACE**



Kiểm tra lời giải trong thời gian đa thức

Nonderterministic Turing machine in **P**olynomial number of step

Cho một tập hợp các số nguyên, tồn tại hay không một tập hợp con có tổng bằng T không?

complement of **NP**

Cho một tập hợp các số nguyên, có phải tất cả tập hợp con đều có tổng khác T không?

Polynomial number of step

Tồn tại lời giải trong thời gian đa thức

Probability

Nondeterministic Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$\delta = Q \times \Sigma \rightarrow [0,1]^{(Q \times \Sigma \times \{L,R\})}$

δ	0	1	#
q_{start}	½: q_L, L; ½: q_R, R	1: q_{accept}	1: q_{reject}
q_L	1: q_L, L	1: q_{accept}	1: q_{reject}
q_R	1: q_R, R	1: q_{accept}	1: q_{reject}

Probability

Lời giải đúng

Chấp nhận lời giải đúng

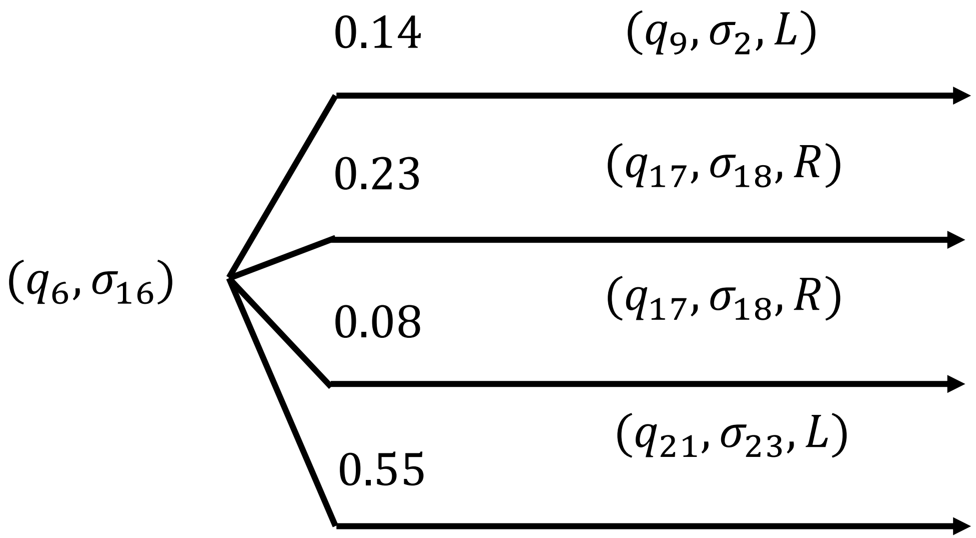
Lớp phức tạp. BPP là tập các vấn đề có thể được giải quyết bằng *PTM* (**P**robabilistic Turing Machine) trong thời gian đa thức (**P**olynomial time) với khả năng xảy ra lỗi. Để chính xác, nếu M là PTM, $L \in \boldsymbol{BPP}$ và x là input, thì:

$$\boxed{x \in L} \Rightarrow \boxed{Prob(M\ accepts\ x)} > \boxed{\frac{2}{3}}$$

Tính hội tụ!

và

$$x \notin L \Rightarrow Prob(M\ rejects\ x) > \frac{2}{3}$$



Lớp phức tạp. RP là tập các vấn đề có thể được giải quyết bằng *PTM* (Probabilistic / **R**andom Turing Machine) trong thời gian đa thức (**P**olynomial time) mà chỉ có khả năng sai tiêu cực. Nói cách khác, nếu M là PTM, $L \in \boldsymbol{RP}$ và x là input thì:

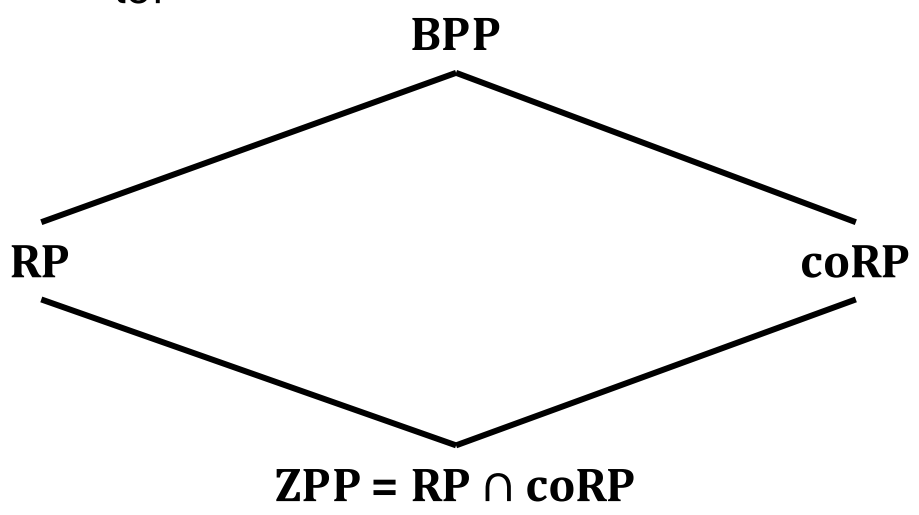
$$x \in L \Rightarrow Prob\ (M\ accepts\ x) > \frac{2}{3}$$

và

$$x \notin L \Rightarrow Prob\ (M\ rejects\ x) = 1$$

Probability

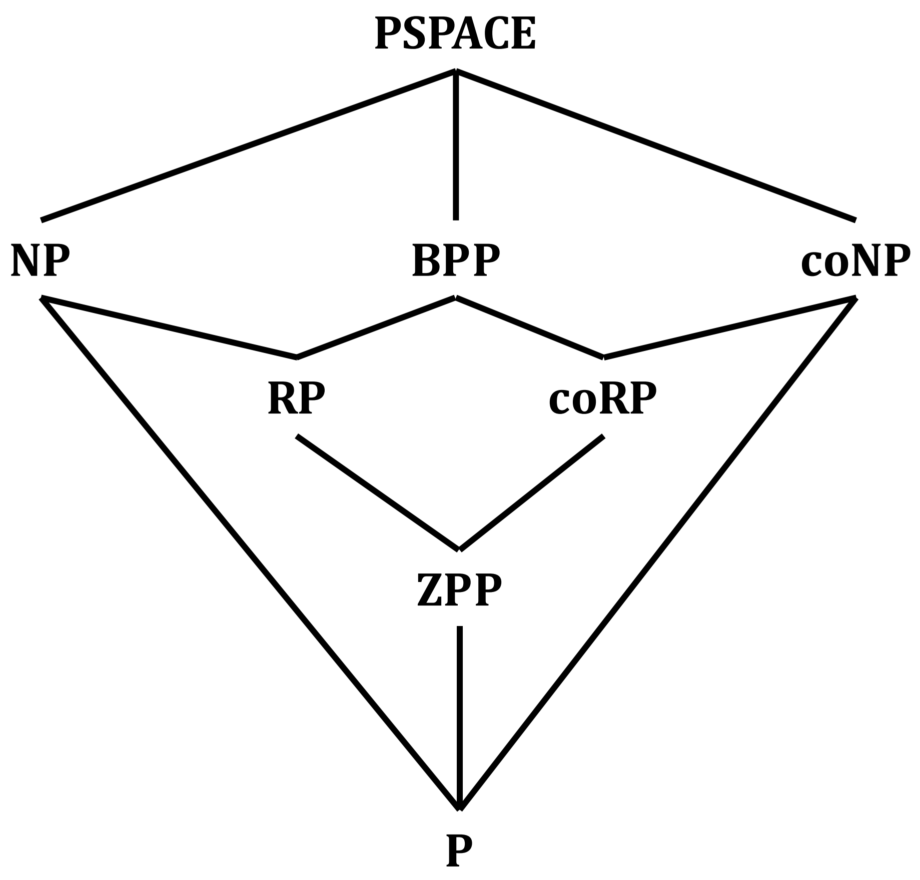
Probabilistic Turing Machine in **P**olynomial time với khả năng xảy ra lỗi



Probabilistic / **R**andom Turing Machine in **P**olynomial time with **false negative** only

Probabilistic / **R**andom Turing Machine in **P**olynomial time with **false positive** only

Probabilistic / Random Turing Machine in **P**olynomial time with **Z**ero error, allow ? result


PSPACE
NP
BPP
coNP
RP
coRP
ZPP
P

Quantum

Quantum Turing machine $M = \{Q, \Sigma, q_{start}, q_{reject}, q_{accept}, \delta\}$

$QTM = \{Q, \Sigma, q_{start}, q_{accept}, q_{reject}, \delta'\}$

$\delta': Q \times \Sigma \times Q \times \Sigma \times \{L, R\} \to \mathbb{C}$

$$(\forall q \in Q, \forall \sigma \in \Sigma) \sum_{q' \in Q,\ \sigma' \in \Sigma,\ D \in \{L,R\}} |\delta'(q, \sigma, q', \sigma', D)|^2 = 1$$

Quantum

δ	0	1	$\#$
q_{start}	$\frac{1}{\sqrt{2}}: q_L, L;$ $\frac{1}{\sqrt{2}}: q_R, R$	$1: q_{accept}$	$1: q_{reject}$
q_L	$1: q_L, L$	$1: q_{accept}$	$1: q_{reject}$
q_R	$1: q_R, R$	$1: q_{accept}$	$1: q_{reject}$

$$|\#000q_{start}0010\#\rangle \mapsto \frac{|\#00q_L00010\#\rangle + |\#0000q_R010\#\rangle}{\sqrt{2}}$$

$$\mapsto \frac{|\#0q_L000010\#\rangle + |\#00000q_R10\#\rangle}{\sqrt{2}}$$

$$\mapsto \frac{|\#q_L0000010\#\rangle + |\#00000q_{accept}10\#\rangle}{\sqrt{2}}$$

Quantum

$$\boldsymbol{EQP \subseteq ZQP \subseteq BQP}$$

Simon, Grover

Lớp phức tạp. BQP là tập các vấn đề có thể được giải quyết bằng QTM (**Q**uantum Turing Machine) trong thời gian đa thức (**P**olynomial time) với sai số (**B**ounded error) bị chặn cả hai biên. Nói cách khác, nếu M là QTM, $L \in BQP$ và x là input, thì

$$x \in L \Rightarrow Prob\ (M\ accepts\ x) > \frac{2}{3}$$

và

$$x \notin L \Rightarrow Prob\ (M\ rejects\ x) > \frac{2}{3}$$

Same as ZPP

Lớp phức tạp. ZQP là tập các vấn đề có thể được giải quyết bằng QTM (**Q**uantum Turing Machine) trong thời gian đa thức (**P**olynomial time) mà không xảy ra lỗi (**Z**ero error). Nói cách khác, nếu M là QTM, $L \in \boldsymbol{ZQP}$ và x là input, thì có một cơ hội nhỏ hơn 50% QTM dừng ở trạng thái "không biết" (do not know), trường hợp ngược lại:

$$x \in L \Rightarrow Prob\ (M\ accepts\ x) = 1$$

và

$$x \notin L \Rightarrow Prob\ (M\ rejects\ x) = 1$$

Deutsch-Jozsa, Require only 1 query

Lớp phức tạp. EQP là tập các vấn đề có thể được giải quyết bằng QTM (**Q**uantum Turing Machine) trong thời gian đa thức chính xác (**P**olynomial time **E**xactly) (không xảy ra lỗi). Nói cách khác, nếu M là QTM, $L \in EQP$ và x là input, thì:

$$x \in L \Rightarrow Prob\ (M\ accepts\ x) = 1$$

Và

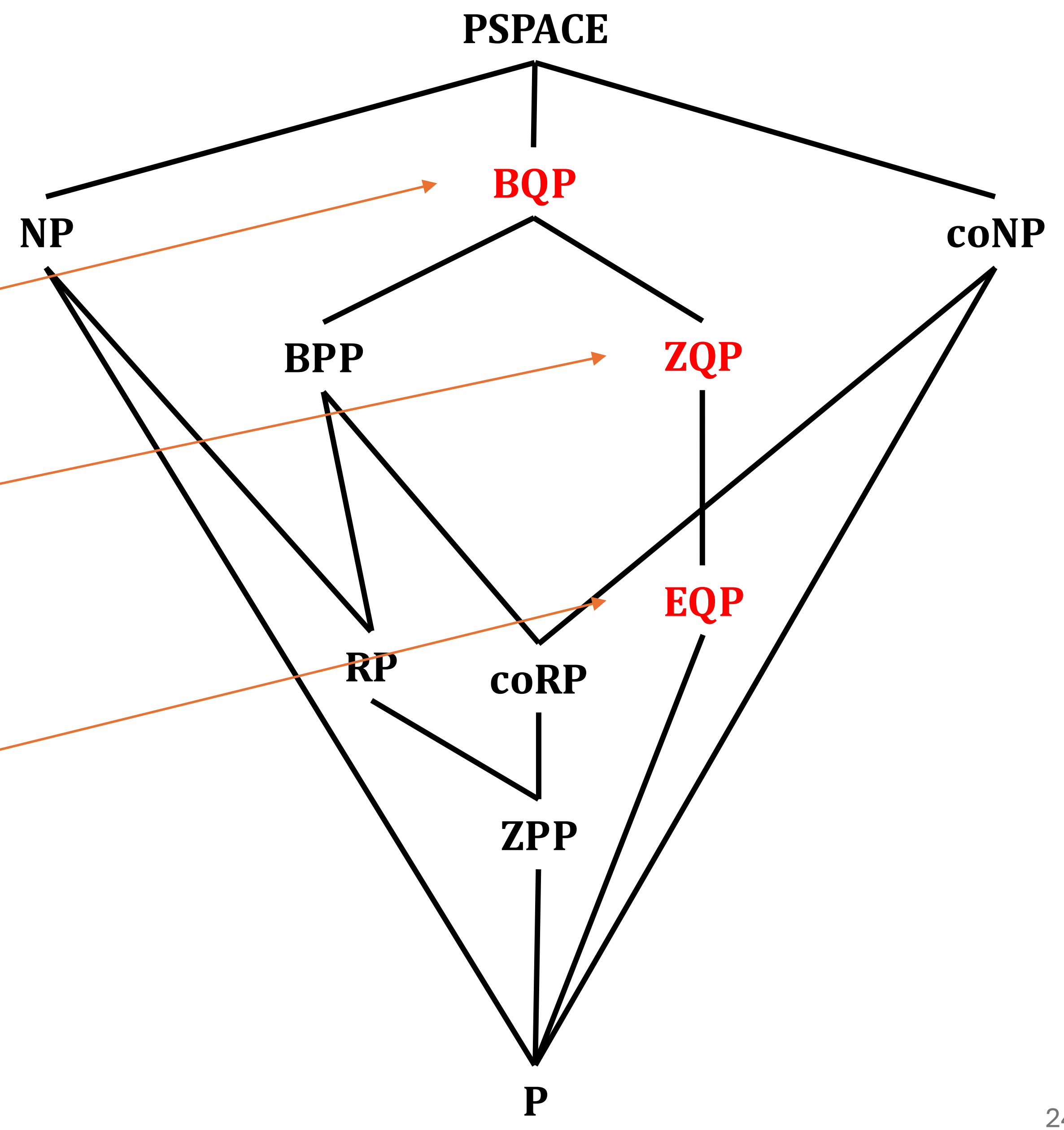
$$x \notin L \Rightarrow Prob\ (M\ rejects\ x) = 1$$

Quantum

Quantum Turing Machine in **P**olynomial time with **B**ounded error (sai số bị chặn)

Quantum Turing Machine in **P**olynomial time with **Z**ero error

Quantum Turing Machine in **P**olynomial **E**xactly time with no error



2. Grover search

Spatial search (discrete time)

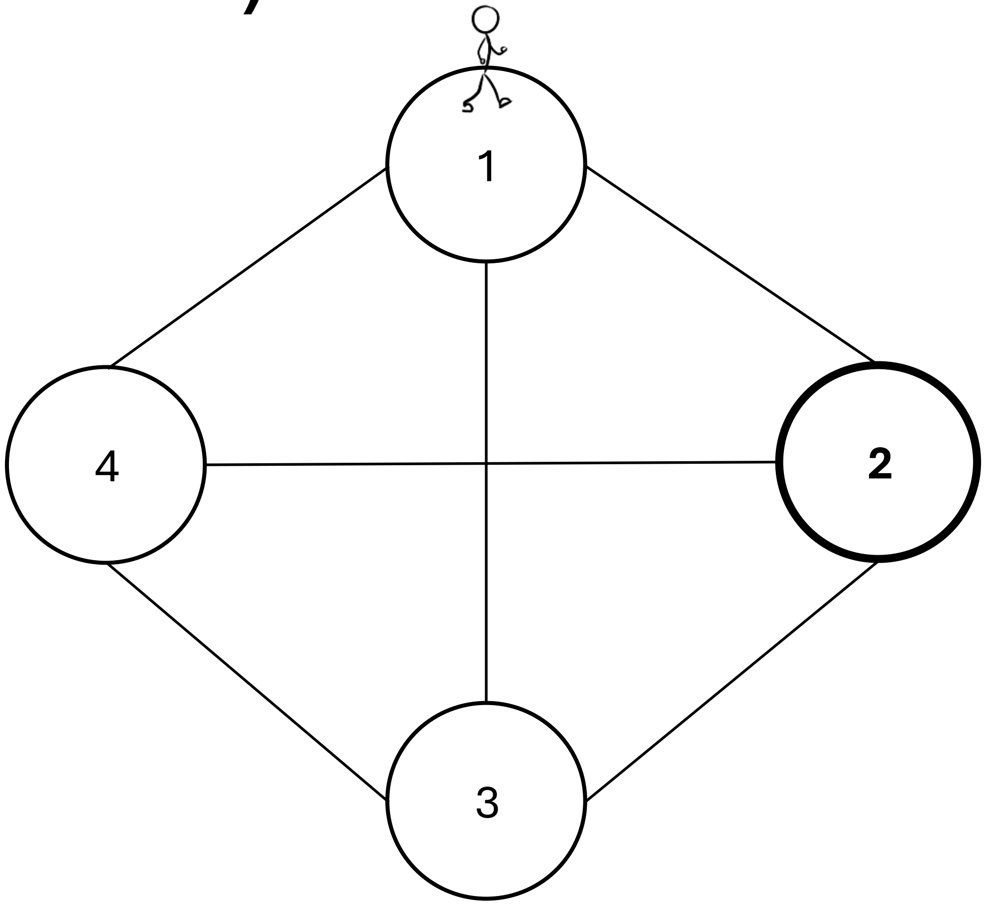
Assume that we have a complete graph with 4 nodes.

One approach is using random walk.

Note that:

Walker must be in somewhere

$$\Rightarrow \sum P_j = 1$$

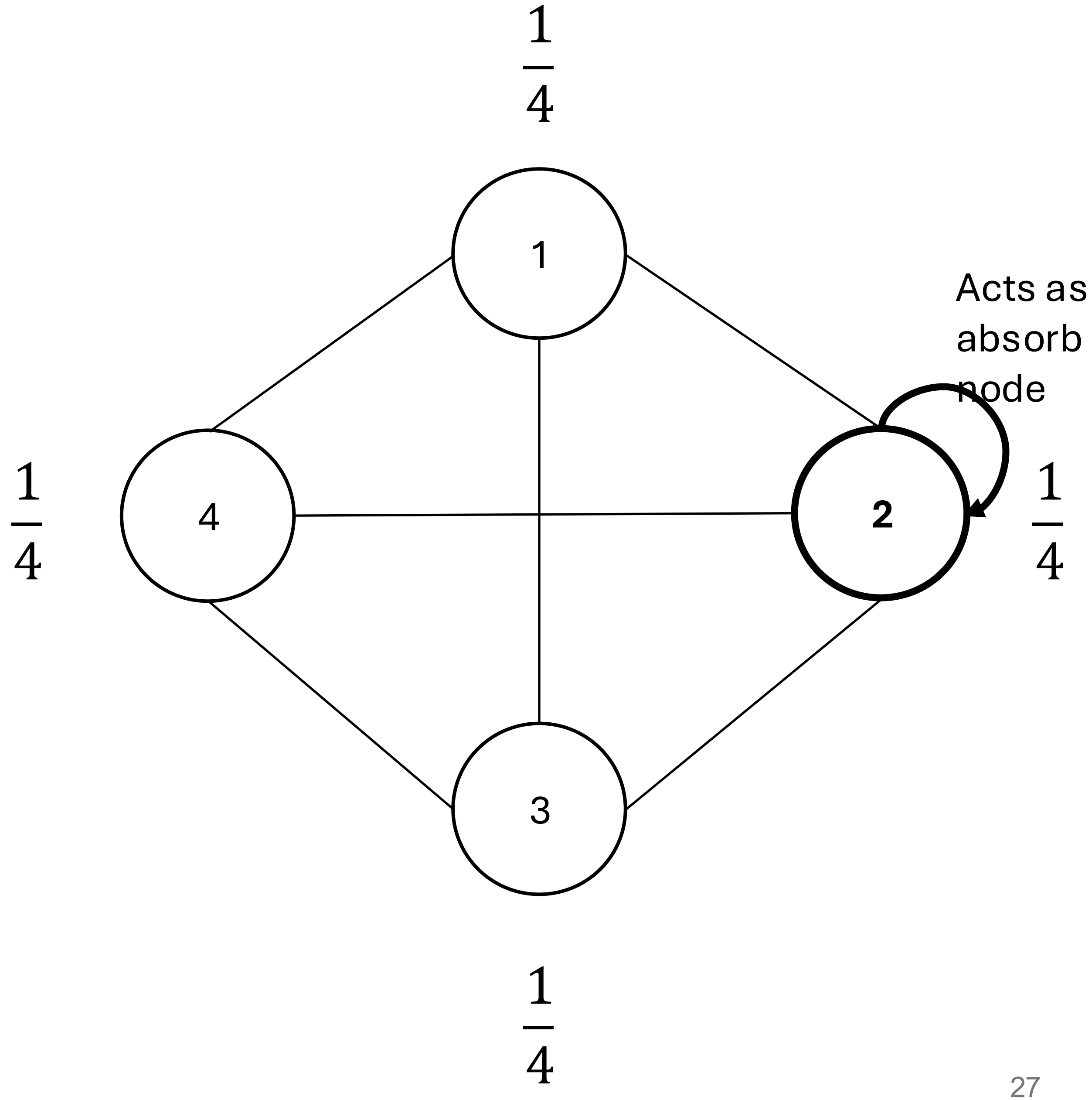


Random walk

Initial state: ¼ at all node

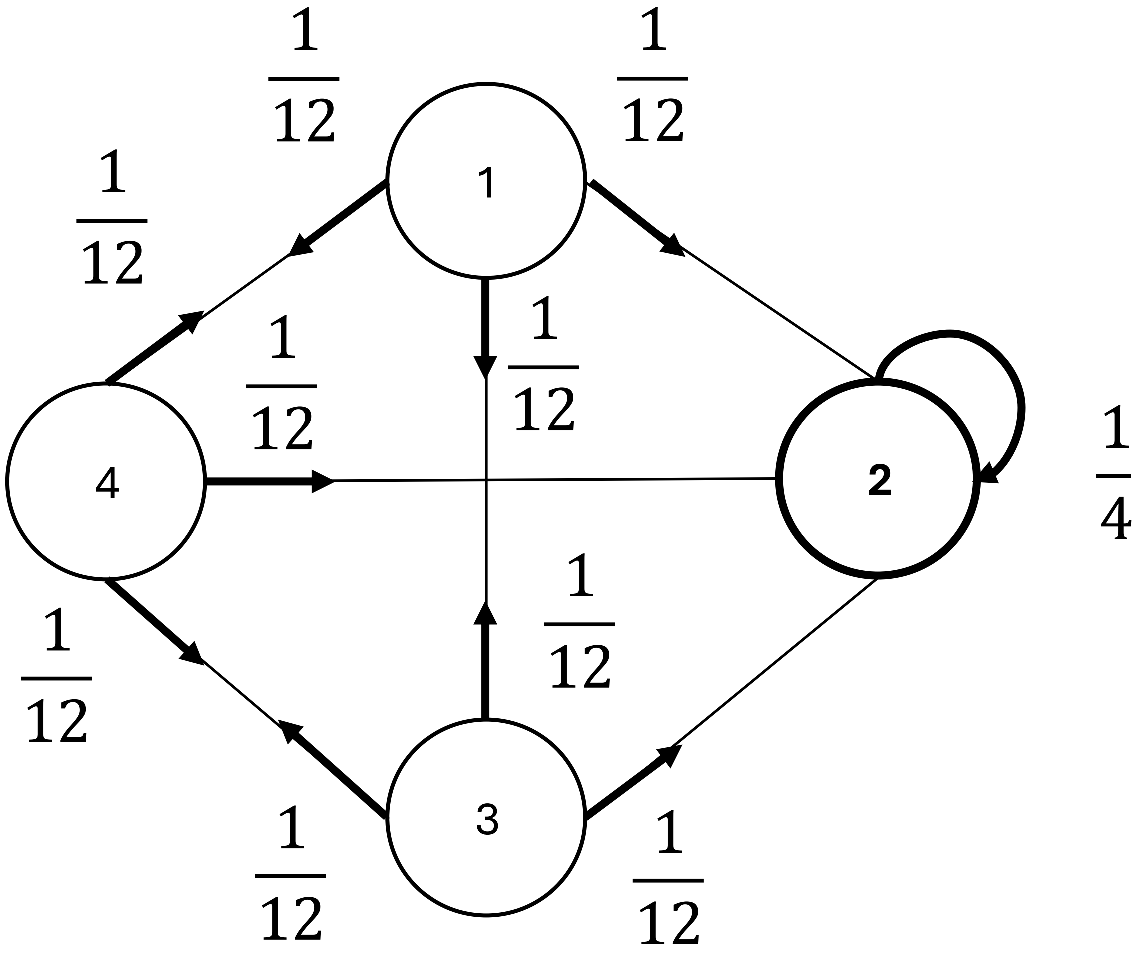
At step k: jump to other random nodes

$$P_1 = P_2 = P_3 = P_4 = 1/4$$



Random walk

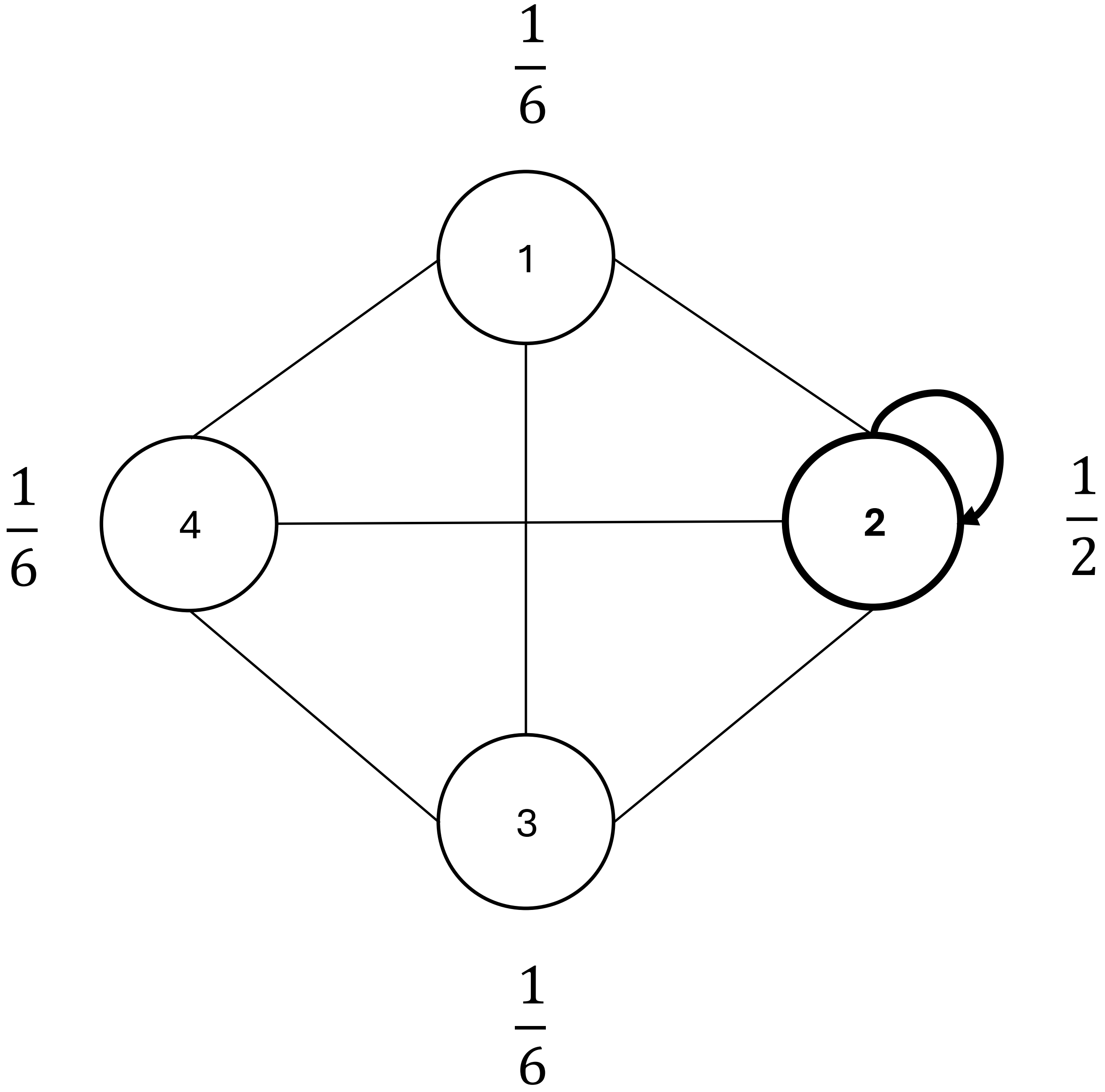
At step 1:



Random walk

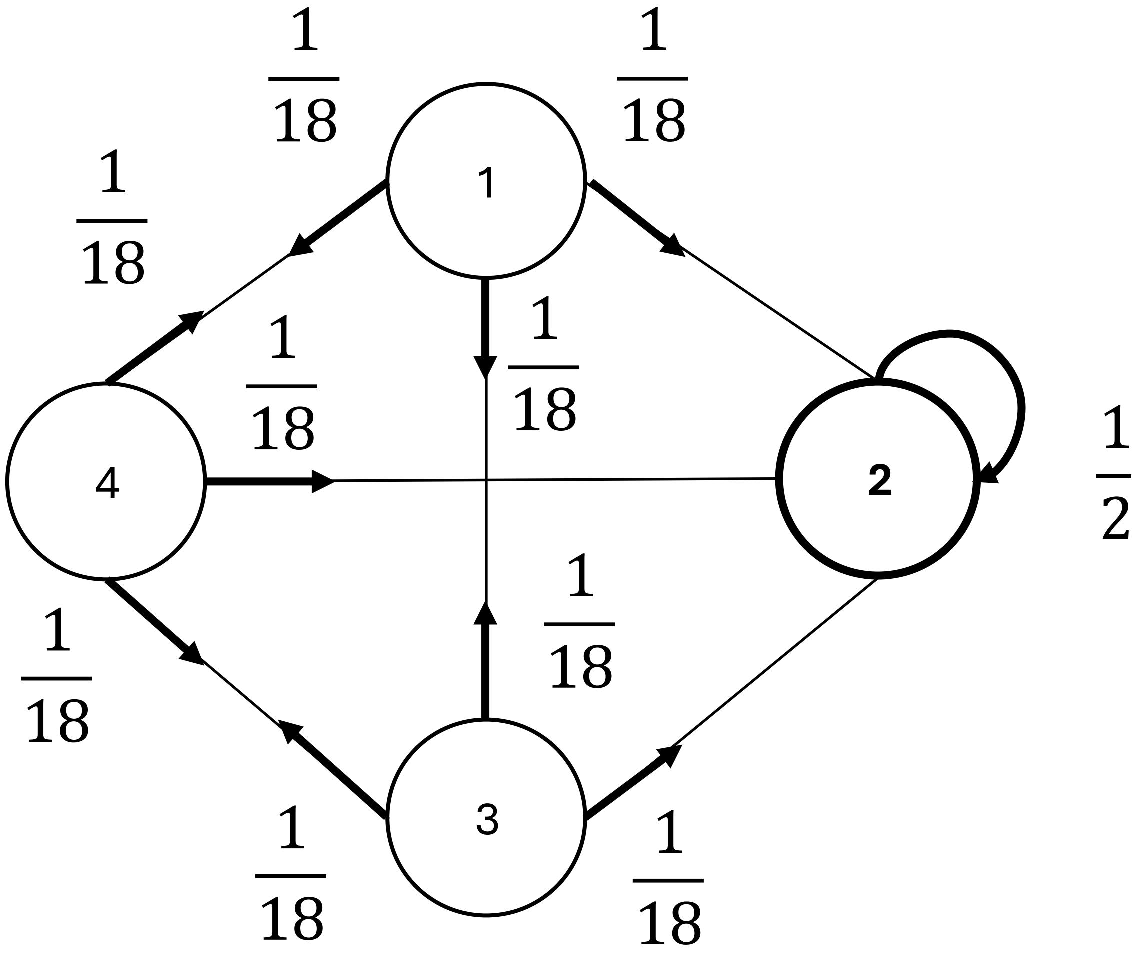
When done step 1:

$$P_1 = P_3 = P_4 = \frac{1}{12} * 2 = \frac{1}{6}$$
$$P_2 = \frac{1}{4} + \frac{1}{12} * 3 = 1/2$$



Random walk

At step 2:

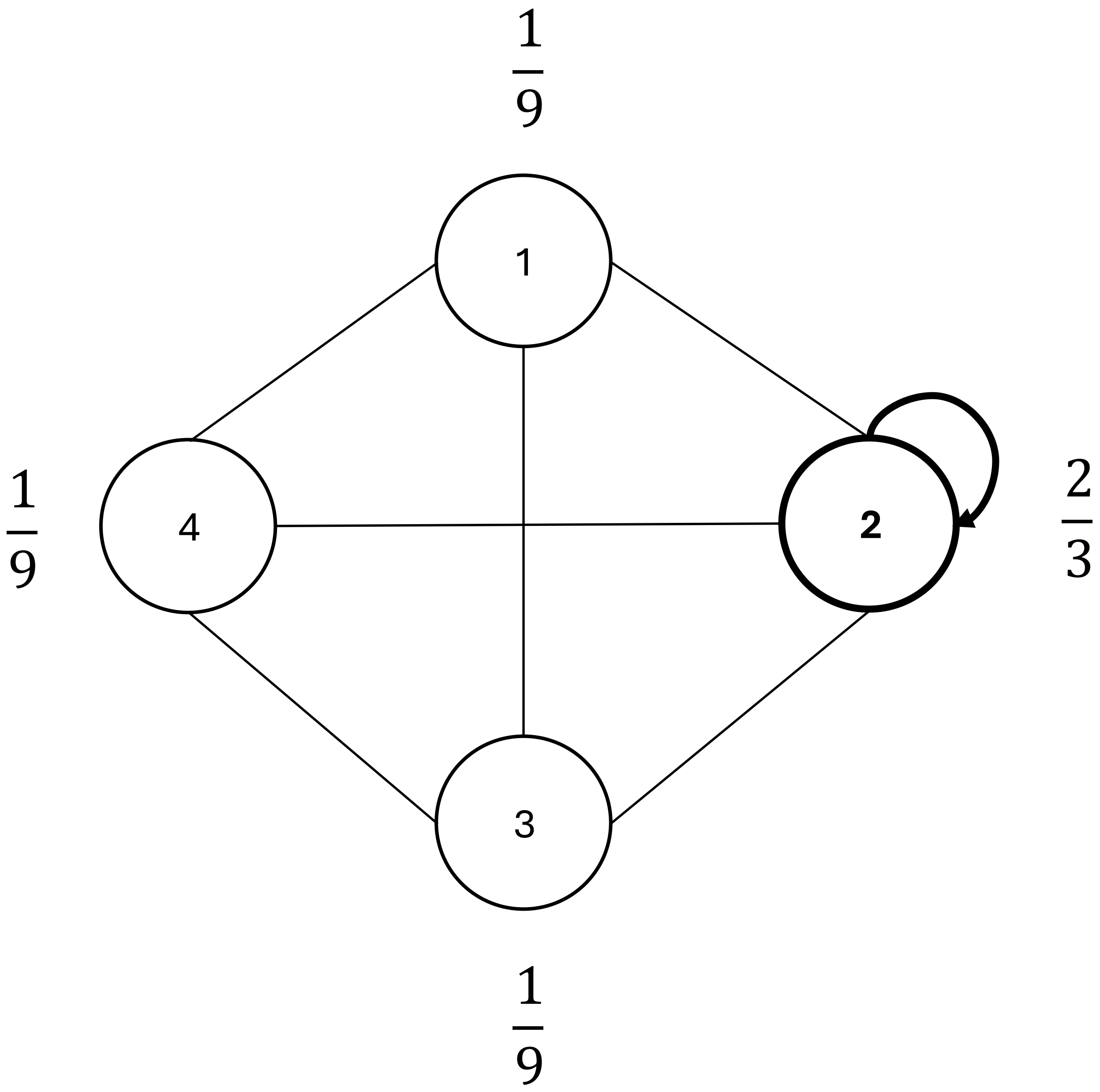


Random walk

When done step 2:

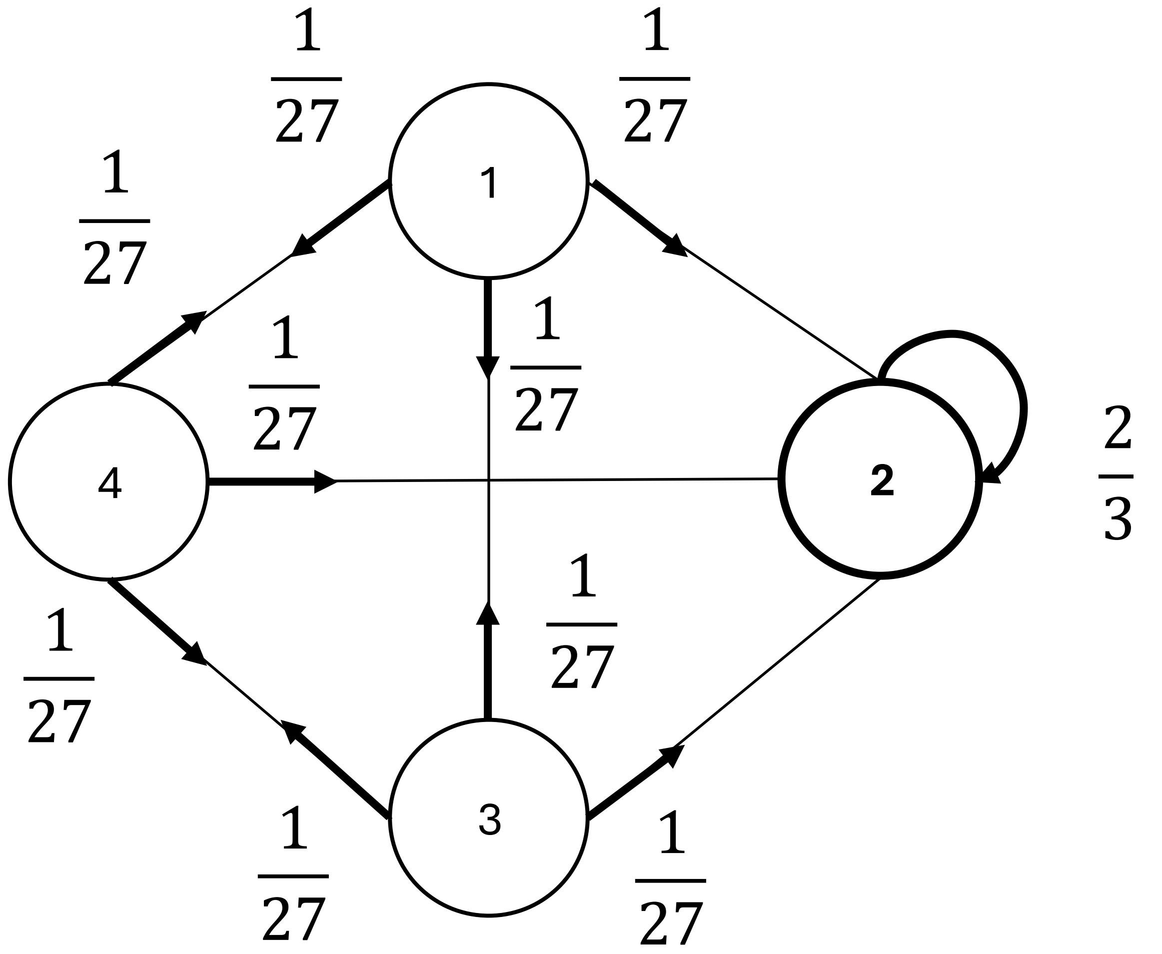
$$P_1 = P_3 = P_4 = \frac{1}{18} * 2 = \frac{1}{9}$$

$$P_2 = \frac{1}{2} + \frac{1}{18} * 3 = \frac{2}{3}$$



Random walk

At step 3:

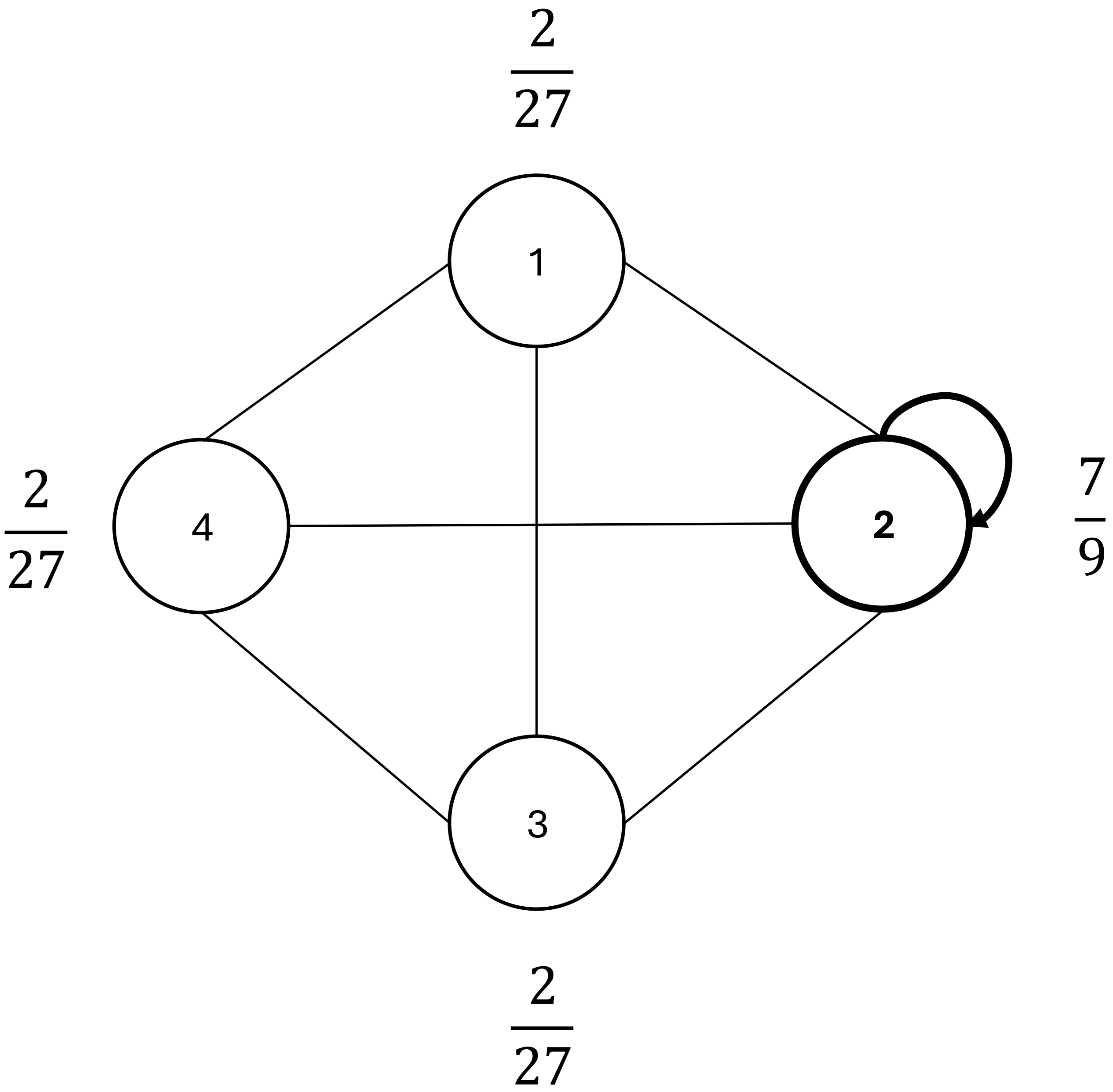


Random walk

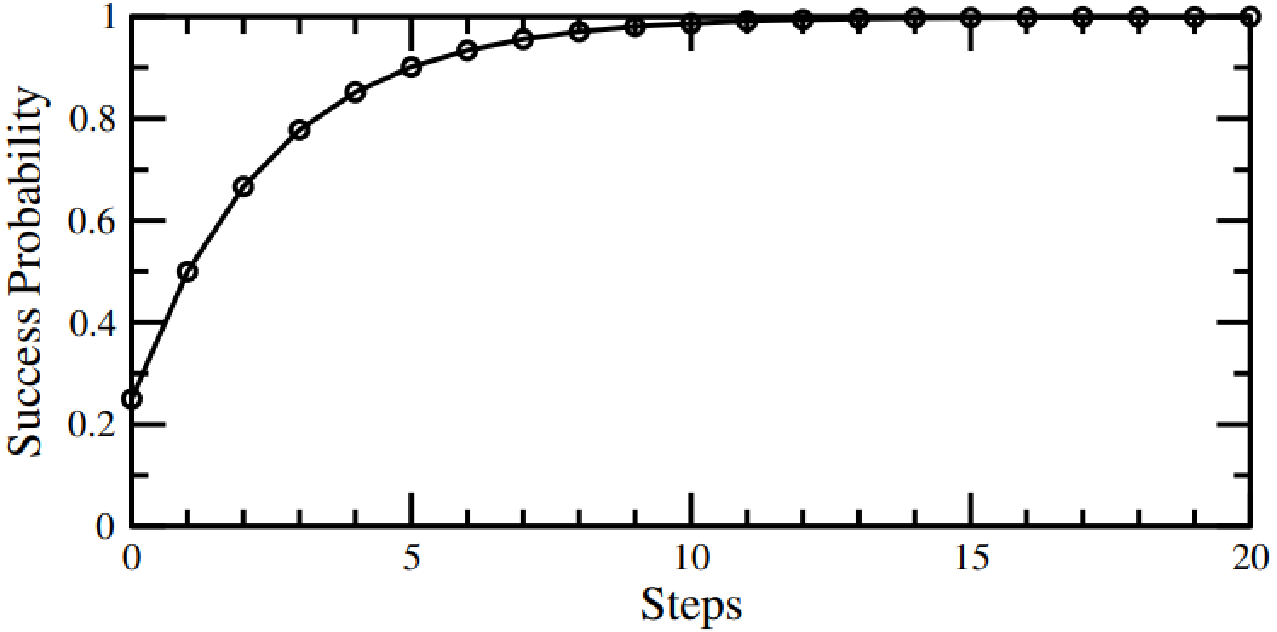
When done step 3:

$$P_1 = P_3 = P_4 = \frac{1}{27} * 2 = \frac{2}{27}$$

$$P(2) = \frac{2}{3} + \frac{1}{27} * 3 = \frac{7}{9}$$



Random walk



Prob(Success) at time t =

$$1 - \frac{N-1}{N}\left(\frac{N-2}{N-1}\right)^t$$

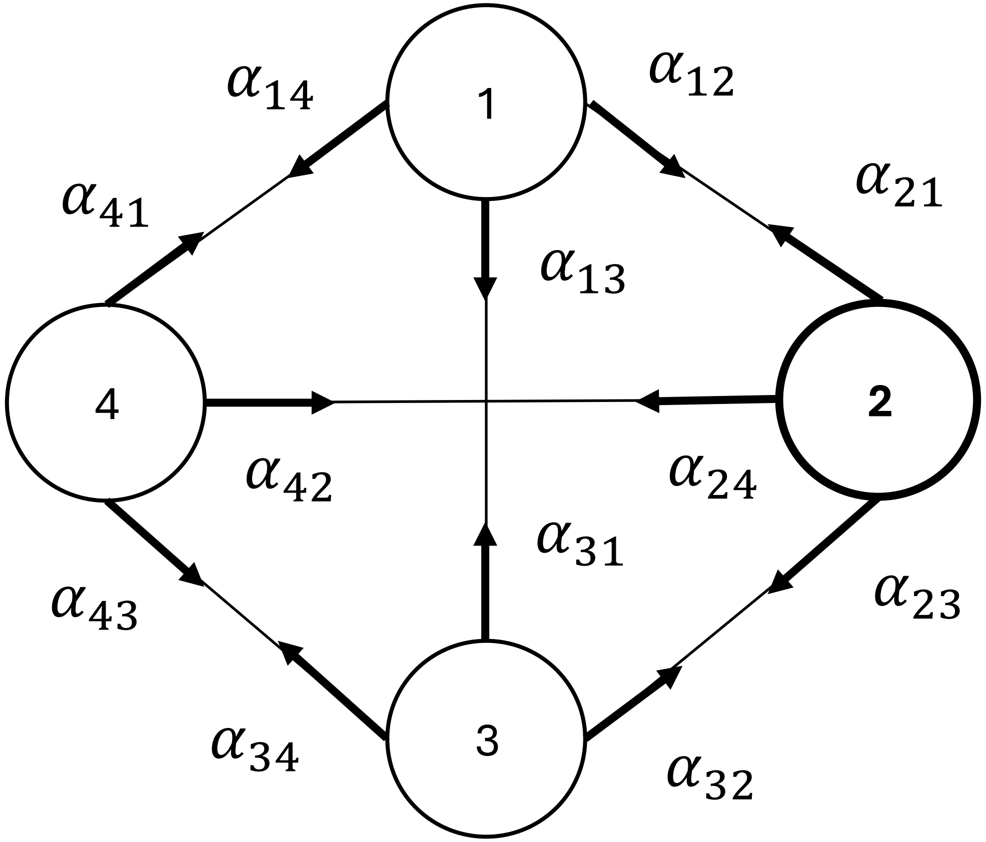
Runtime: $O(N)$

Quantum walk

Change 4 probs to 12 amplitudes of a wave function $|\psi\rangle \coloneqq \{\alpha_i\}$

Init: $|\psi(0)\rangle$

Note that at intermediate steps we cannot observe the system. Then we do not know the current node is the marked node or not.



At node 2: A = $\frac{-1}{\sqrt{12}} * \frac{3}{3}$

Quantum walk

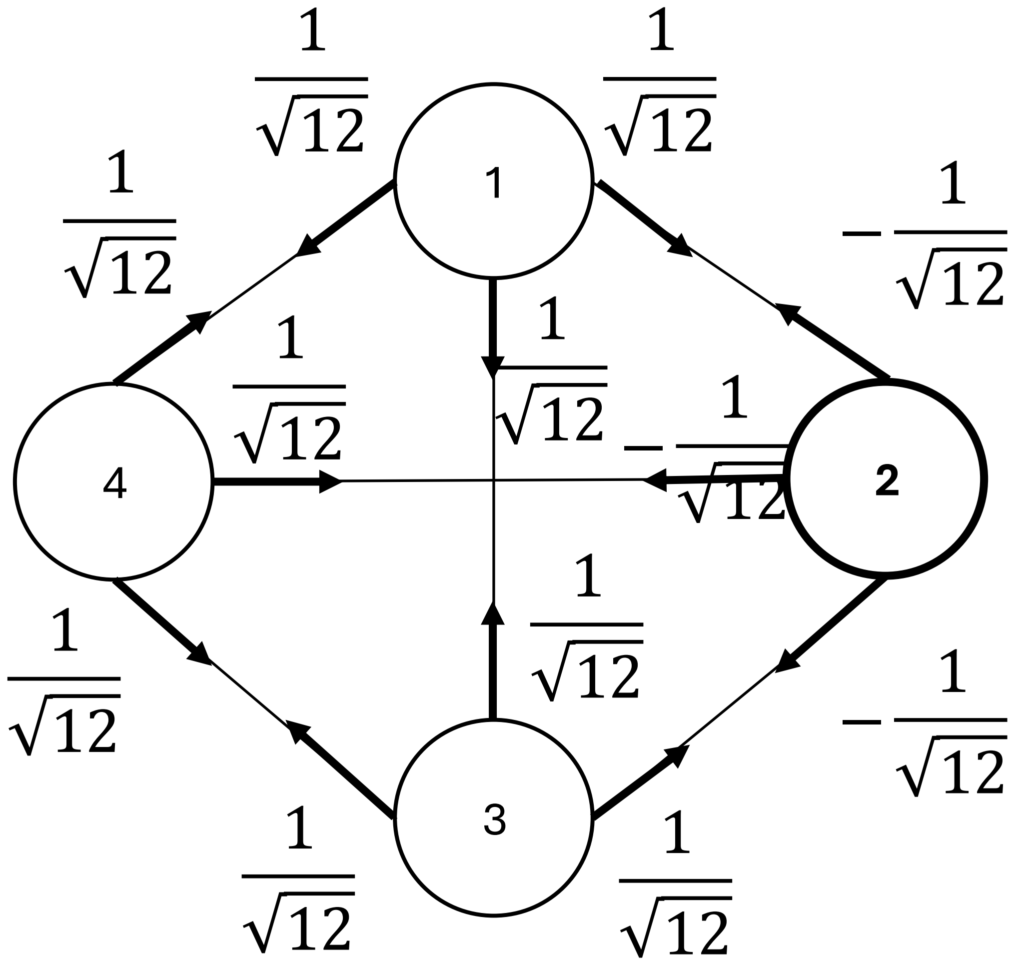
At step 1

1a. Query oracle:

$$Q|\psi(0)\rangle$$

1b. Apply a quantum coin flip+shift (-I+2A) that inverts each amplitude about the average amplitude at its vertex. (Nothing happen)

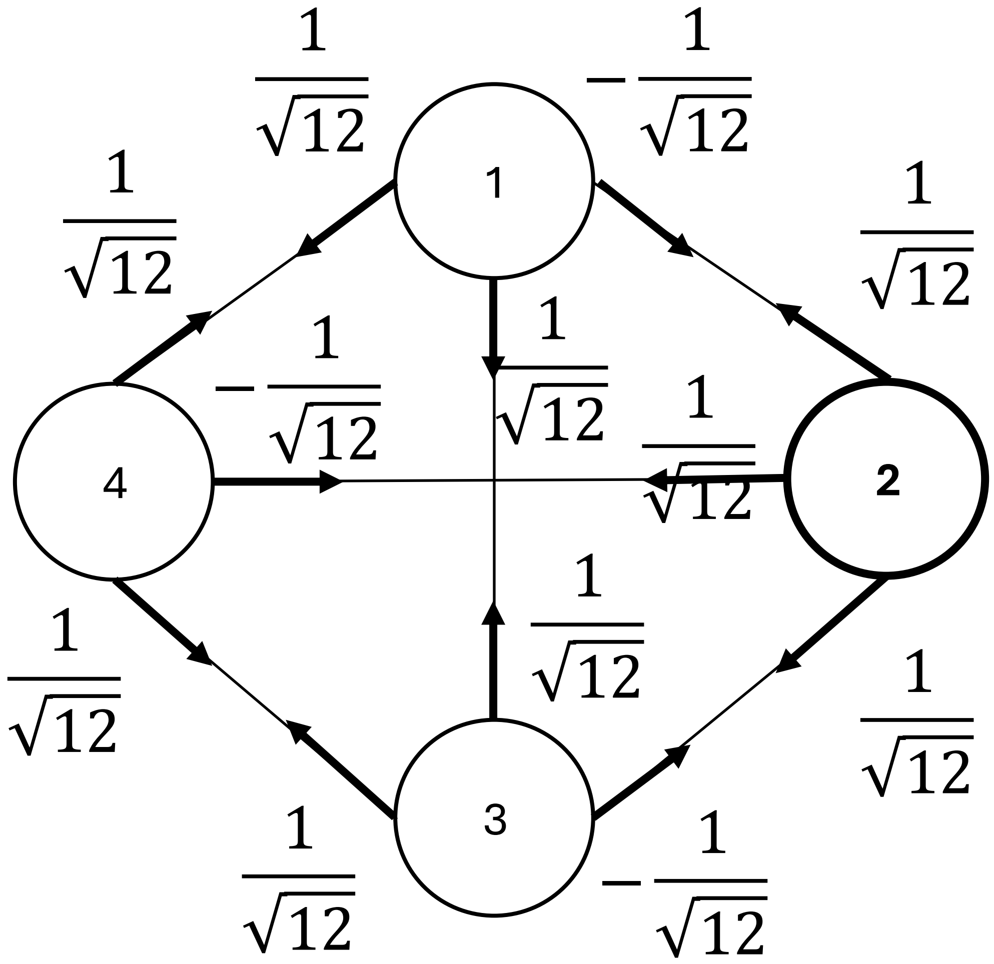
$$C(Q|\psi(0)\rangle)$$



Quantum walk

1c. Shift operator (walk): walker at vertex i pointing to vertex j to hop to vertex j and point back to vertex i ⇒ Swap α_{ij} with α_{ji}.

$$SCQ|\psi(0)\rangle = |\psi(1)\rangle$$

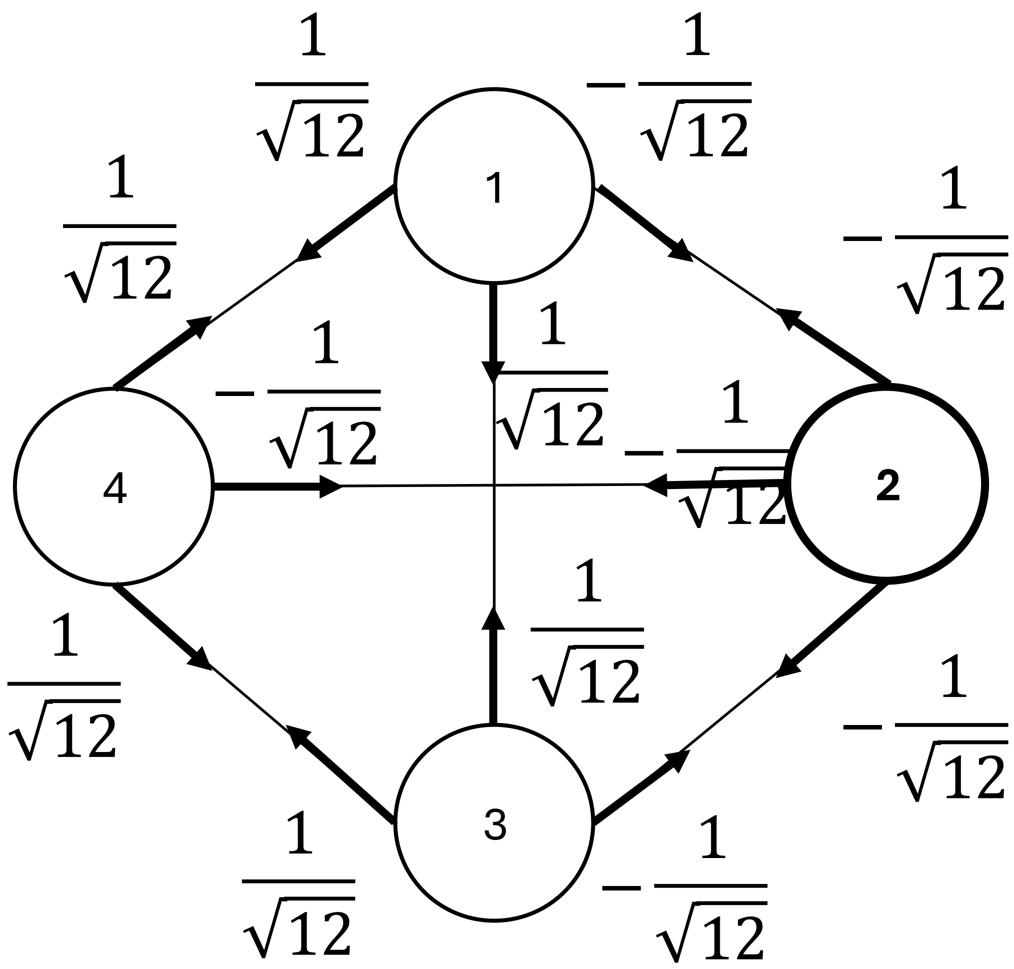


Quantum walk

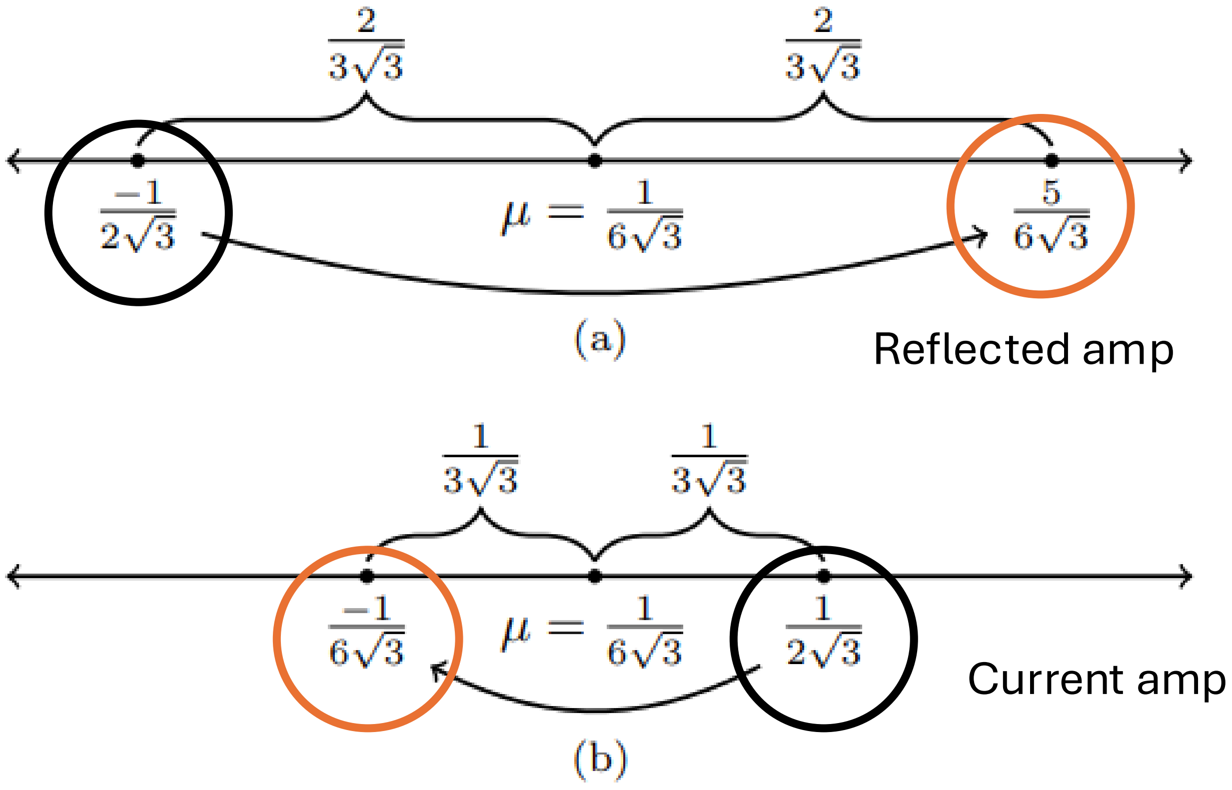
At step 2

2a. Query oracle:

$$Q|\psi(1)\rangle$$



Quantum walk



2b. The coin inverts or reflects each amplitude about their average.

(-I+2A)

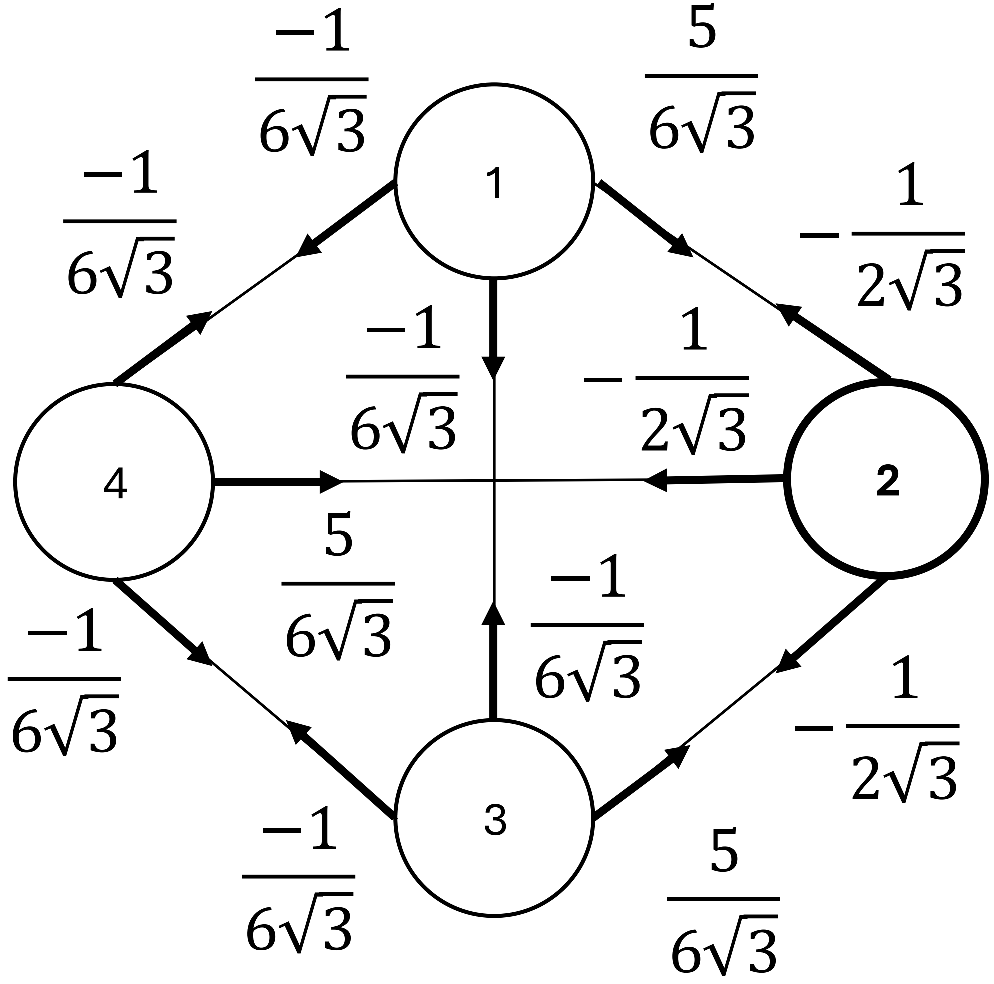
At node 1: $A = \frac{\frac{1}{\sqrt{12}}+\frac{1}{\sqrt{12}}-\frac{1}{\sqrt{12}}}{3} = \frac{1}{6\sqrt{3}}$

$$\Rightarrow \alpha_{12} = -\alpha_{12} + 2A = \frac{5}{6\sqrt{3}}$$

$$\Rightarrow \alpha_{13} = -\alpha_{13} + 2A = \frac{-1}{6\sqrt{3}}$$

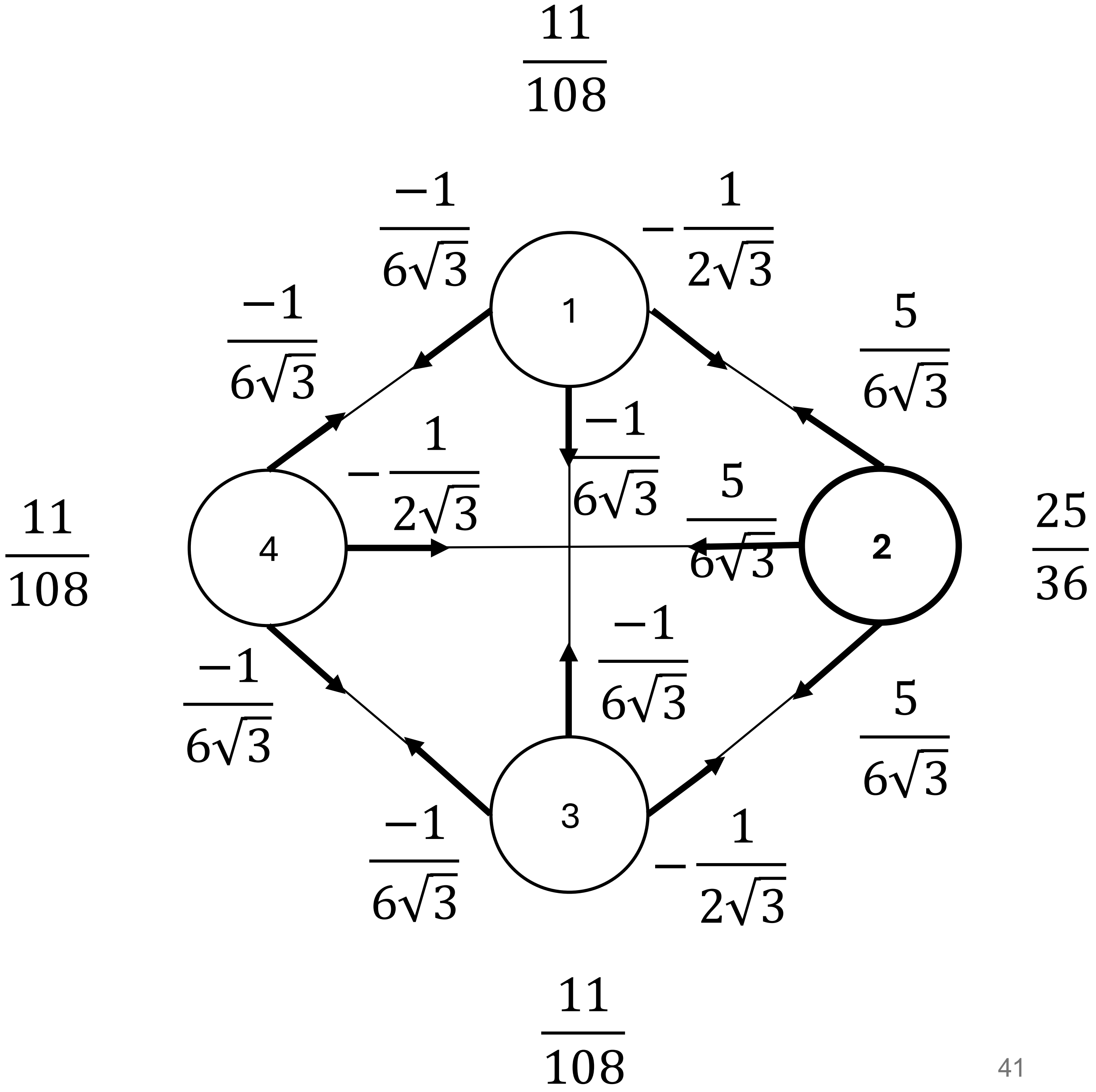
Quantum walk

2c. Shift operator (walk): swap α_{ij} with α_{ji}.

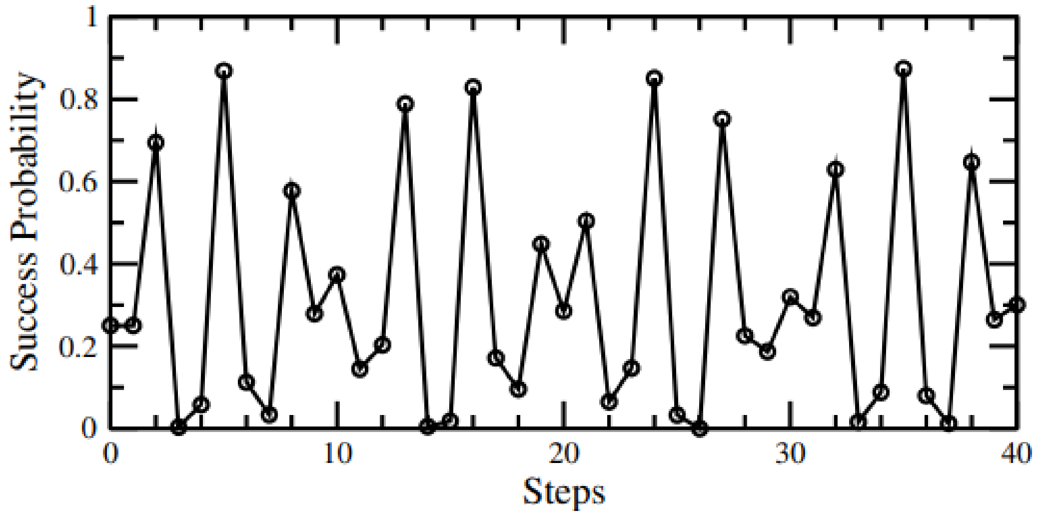


Quantum walk

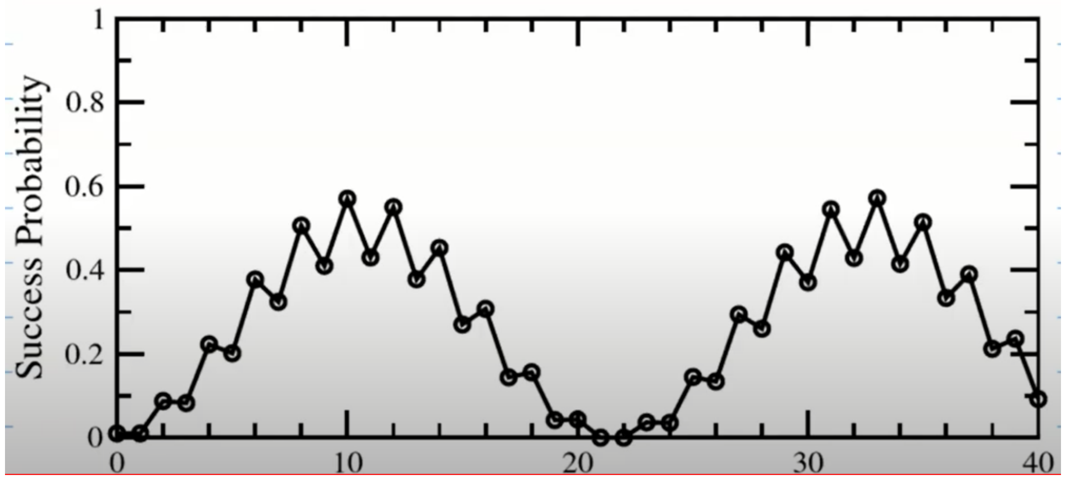
2c. Shift operator (walk): swap α_{ij} with α_{ji}.



Quantum walk



N=4



N=10
0

Prob(Success) at time t =

$$\left\{(N-1)\left[\cos(\phi t)+\sqrt{2N-3}\,\sin(\phi t)\right]+(-1)^t(N-2)\right\}^2/\left[(2N-3)^2 N\right],$$

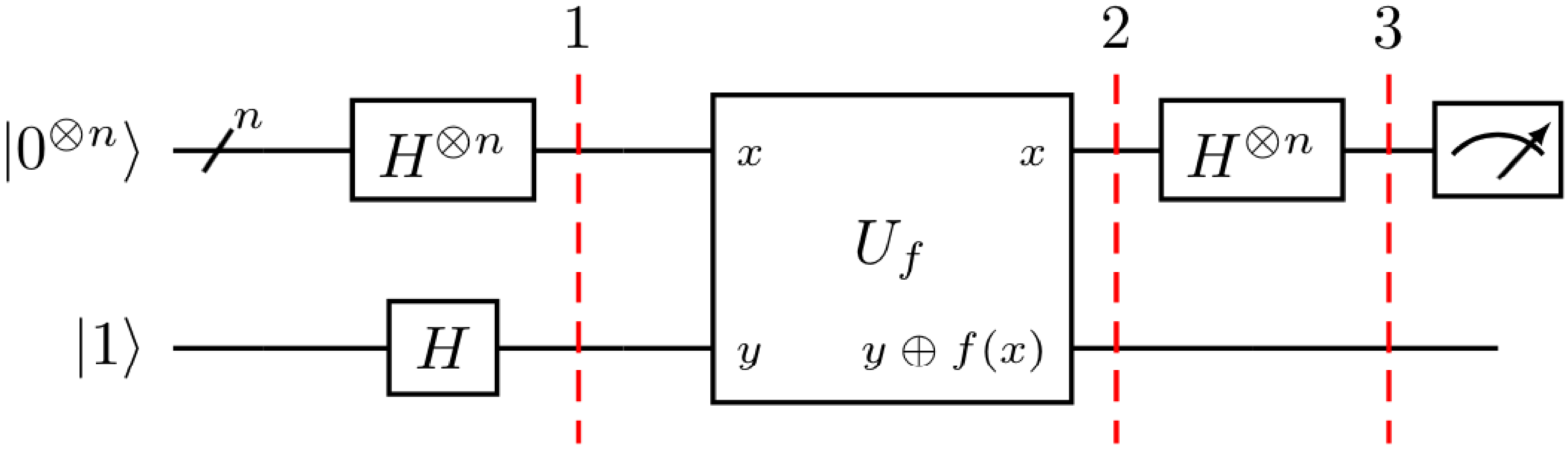
$$\text{where } \phi=\sin^{-1}\left(\frac{\sqrt{2N-3}}{N-1}\right)$$

Runtime: $O(\sqrt{N})$

3. Sources of quantum advantage

Largely it comes from superposition.

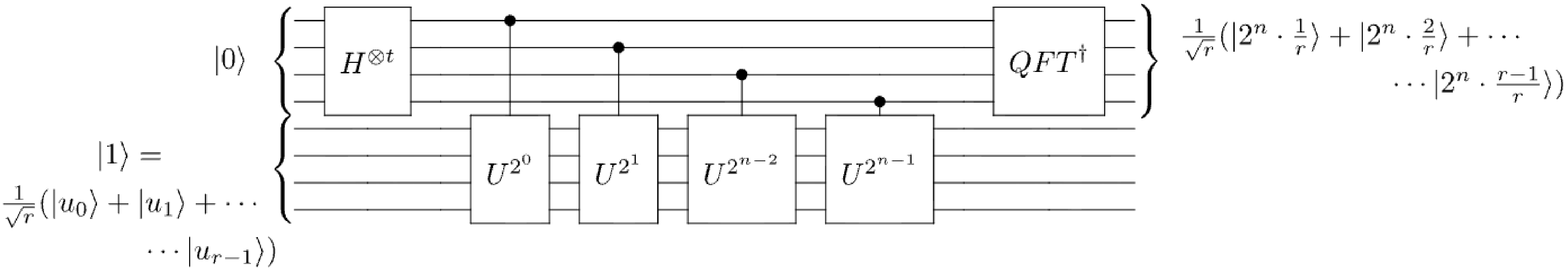
Look Deutsh-Josza algorithm



3. Sources of quantum advantage

Largely it comes from superposition.

Look Shor’s algorithm



3. Sources of quantum advantage

Total runtime = $T_{prepare}$)build a useful representation) + T_{solve} (find the solution).

Recent quantum algorithms: $T_{prepare} \gg T_{solve}$

The right benchmark is the strongest available classical workflow, heuristics included - not an idealized worst case.

Q: “Can quantum give speedups under the same heuristic assumptions classical methods use?”

3. Source of Advantage: example in quantum simulation

Classical

- Must store d^n complex amplitudes explicitly.
- Matrix–vector multiplication scales as $O(d^{2n})$.

- **Quantum**
  - Represents the same Hilbert space with only n qubits.
  - Under efficient block encodings, cost can scale as $O(poly(n))$; storage becomes $O(n)$.

3. Source of Advantage: locality and light cone

We usually solving the eigenvalue problem:

$$\hat{H}c = Ec$$

1. $\hat{H}$ is sparsity => ok, solvable on classical computer
2. But $\hat{H}^k$ generate a light-cone structure. We must to use resources for saving correlation information
   $L_c \sim nR_c$,where n is the number of iterations and R_c denotes the interaction range. Consequently, $N_{\text{corr}} \sim (nR_c)^{mD} = L_c^{mD}$ where D is the physical dimension of the system.

   The resulting quantum advantage is approximately $\frac{T_C}{T_Q} \sim \frac{L_C^{mD}}{T_{BE}}$,

Bottleneck: Data Preparation

1. Classical data must first be converted into quantum state
2. A N – dim classical vector may need only $O(\log N)$ qubits.
3. Without QRAM, state-preparation cost can scale badly enough to eliminate any theoretical speedup entirely.
4. Block encoding: QSVT and Hamiltonian simulation assume a unitary that embeds a matrix — constructing it is often non-trivial and costly for dense matrices.

PIAS
PHENIKAA INSTITUTE FOR ADVANCED STUDY

Quantum Machine Learning in Multi-Qubit Phase-Space

Speaker: Timothy Heightman
The Institute of Photonic Sciences, ICFO
Barcelona, Spain

10:00 am – 11:00 am (Hanoi time)
May 25, 2026
PIAS, 24th Floor, Building A9, Phenikaa

Quantum machine learning (QML) seeks to exploit the intrinsic properties of quantum mechanical systems, including superposition, coherence, and quantum entanglement for classical data processing. However, due to the exponential growth of the Hilbert space, QML faces practical limits in classical simulations with the state-vector representation of quantum system. On the other hand, phase-space methods offer an alternative by encoding quantum states as quasi-probability functions. Building on prior work in qubit phase-space and the Stratonovich-Weyl (SW) correspondence, we construct a closed, composable dynamical formalism for one- and many-qubit systems in phase-space. This formalism replaces the operator algebra of the Pauli group with function dynamics on symplectic manifolds, and recasts the curse of dimensionality in terms of harmonic support on a domain that scales linearly with the number of qubits. It opens a new route for QML based on variational modelling over phase-space.

CMP Seminar

physics @VNUHCM-US
IFAS INSTITUTE OF FUNDAMENTAL AND APPLIED SCIENCES
Quantum research Ho Chi Minh city

ONLINE WEBINAR

QUANTUM TALK #4

TOPIC: Toward Scalable Quantum Machine Learning in Variational and Non-Variational Frameworks

Dr. Thiparat Chotibut
Chulalongkorn University (Thailand)
Moving to NTU (Singapore) in November 2026

08:00 PM, June 15th, 2026 (VN time)

Duration: **40 Minutes** / Language: **English**

In this talk, we address the scalability challenges of quantum machine learning algorithms by examining both variational and non-variational frameworks. We begin by analyzing variational quantum algorithms, which, while useful at small system sizes, often suffer from untrainability at scale due to barren plateau phenomena. We discuss several warm-start strategies to partially bypass these limitations. Beyond variational methods, we explore quantum reservoir computing and quantum extreme learning machines, which offer alternative learning paradigms by utilizing fixed (i.e., non-trainable) quantum dynamics to generate high-dimensional feature spaces. We outline how such non-variational approaches should be designed so that scaling up the system size does not suffer from concentration-of-measure phenomena. Lastly, we provide a forward-looking perspective on how scalable quantum machine learning may progress as hardware and algorithmic strategies co-evolve, culminating in autonomous (self-training) quantum machine learning proposal.

Registration

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Quantum research Ho Chi Minh city

ONLINE WEBINAR

QUANTUM TALK #6

TOPIC: Quantum machine learning and the journey toward scalability

Dr. Supanut Thanasilp
Chulalongkorn University (Thailand)

08:00 PM, August 4th, 2026 (VN time)

Duration: **60 Minutes** / Language: **English**

Quantum systems are well known for their ability to generate non-classical patterns. The idea that they could also be used to recognize highly complex patterns hidden in data is deeply exciting, giving rise to the young interdisciplinary field of quantum machine learning (QML). However, while quantum advantage in data analysis may in principle emerge from the exponentially large Hilbert space of quantum systems, the scalability of QML models remains under active debate. The very same high-dimensional space that enables expressive learning models can also hinder scalability if not handled carefully. In this talk, I will provide an overview of QML and discuss several central challenges surrounding scalability, including barren plateaus, exponential concentration, and classical simulability, as well as potential opportunities such as warm-start approaches

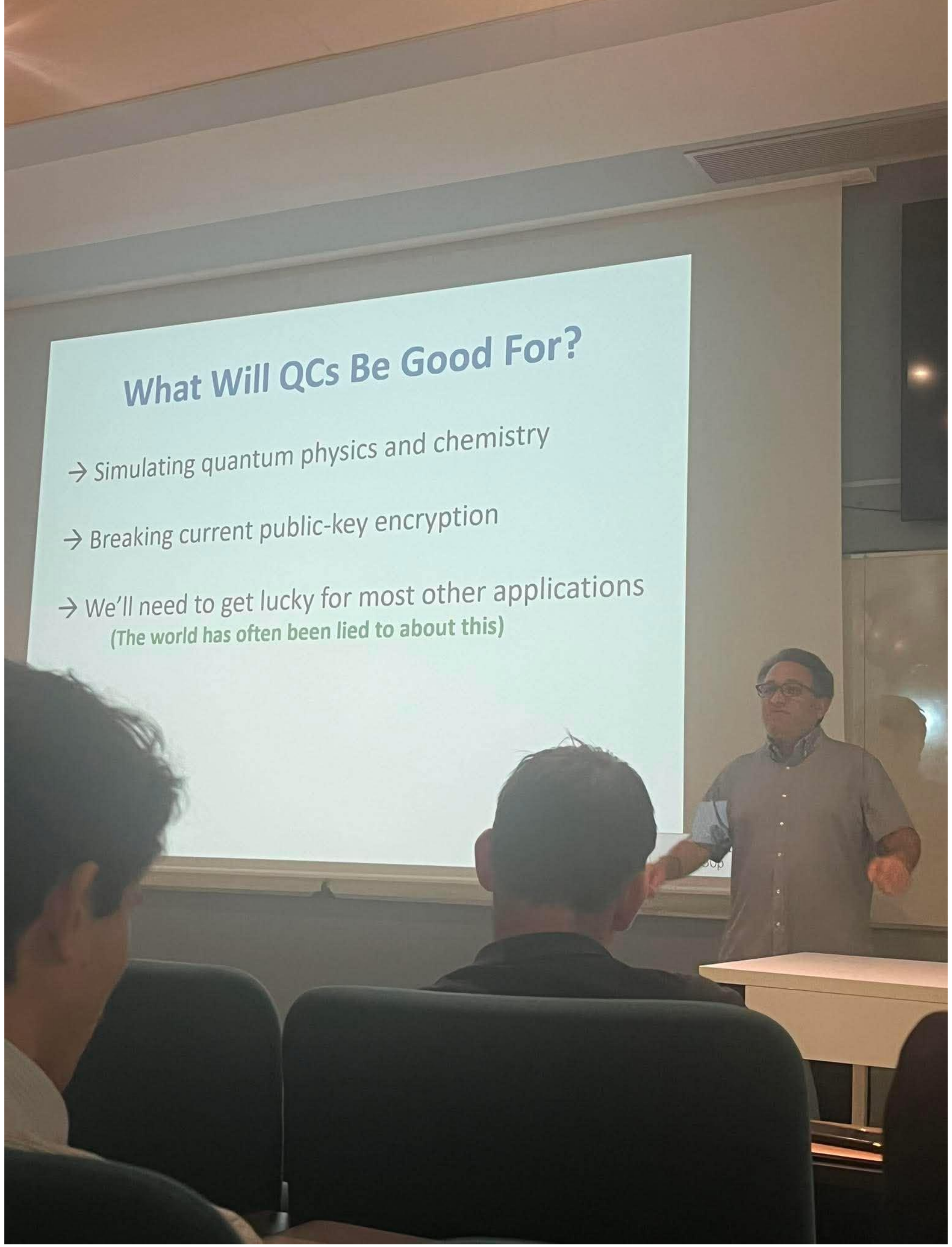
Registration

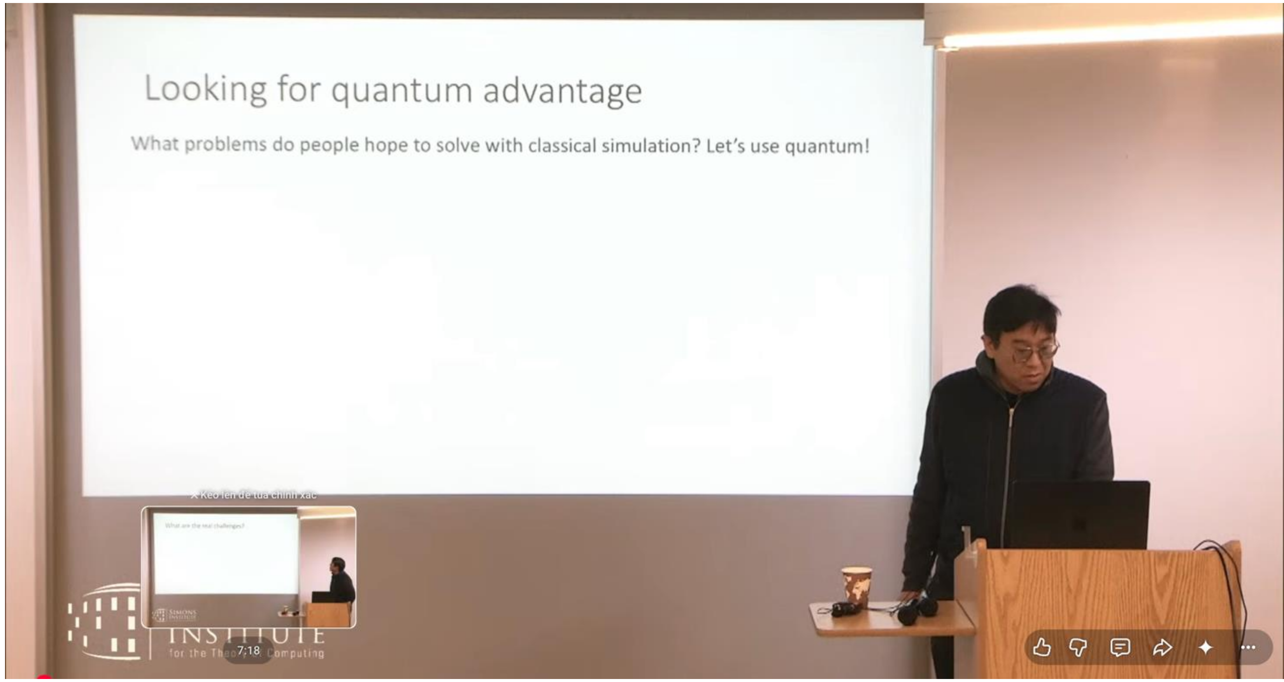
4. Limitations of Quantum Machine Learning

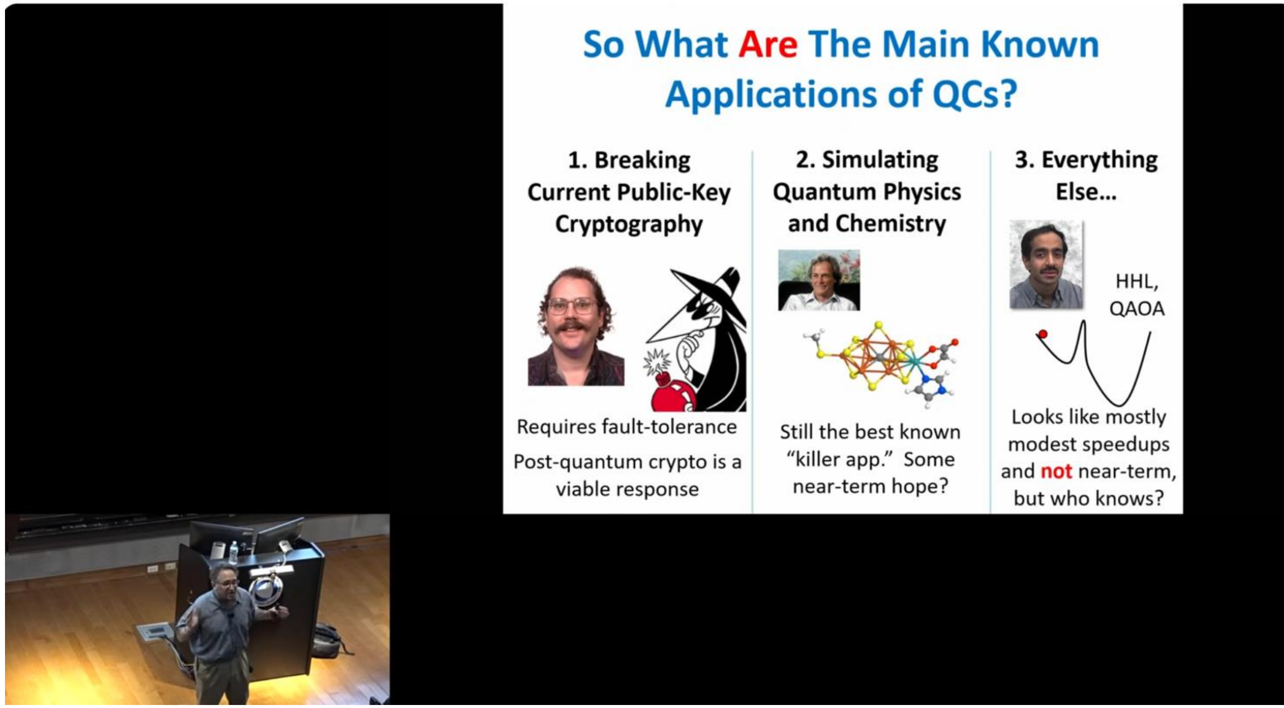
1. QML is one of the most <u>actively studied topics</u>, with many claimed advantages over classical ML.
2. Small-scale evidence: reported gains often come from small experiments or idealized simulations, not current noisy hardware.
3. Hidden data cost: many claimed speedups neglect the cost of data preparation, building the block encoding can cost more than solving the problem classically.
4. Idealized access: complexities like O(poly log n) frequently assume access models that hide real data-acquisition and encoding costs.

Which problems do quantum computers solve efficiently?

1. **A quantum speedup must be judged end-to-end — not by the quantum portion alone.**
2. Quantum simulation, condensed matter physics, and quantum chemistry offer the most near-term opportunities.
3. For QML, the bottleneck is typically converting classical data into quantum-ready form, not the circuit itself.


What Will QCs Be Good For?
→ Simulating quantum physics and chemistry
→ Breaking current public-key encryption
→ We'll need to get lucky for most other applications
(The world has often been lied to about this)


Looking for quantum advantage
What problems do people hope to solve with classical simulation? Let's use quantum!


So What Are The Main Known Applications of QCs?
1. Breaking Current Public-Key Cryptography
Requires fault-tolerance
Post-quantum crypto is a viable response
2. Simulating Quantum Physics and Chemistry
Still the best known "killer app." Some near-term hope?
3. Everything Else...
HHL, QAOA
Looks like mostly modest speedups and not near-term, but who knows?